

Chapter 25 – Geotechnical Instrumentation and Monitoring

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25.1- Introduction

Geotechnical design embraces uncertainty. That uncertainty is assessed during the construction and operational phase of project using various instruments. This chapter covers the different types of geotechnical instruments commonly used for monitoring the short-term and long-term behaviour of the ground supporting the foundations of man-made structures. The main geotechnical parameters for this monitoring are:

- Groundwater pressure measured using piezometers.
- Displacements measured at the ground surface and at depth. Displacements are measured as x,y,z components and are either relative or absolute. Displacements are measured using extensometers, settlement gauges, inclinometers, tiltmeters or change detection using Geodetic measurement methods.
- Strain measured on the surfaces of the ground or structure using strain gauges. Strain gauges can also be imbedded in structures
- Loads measured to quantify the applied load transferred from a structural element to the ground. Frequently measured using load cells and applied in tie-back anchored walls and testing piles.
- Total pressure measured using hydraulic total pressure cells to quantify the contact pressure between the ground and structure.
- Temperature of the ground surface or ground temperature profile with depth, measured using thermistors.

This chapter reviews the various instruments used to measure these parameters and describes data collection methods and acquisition systems.

Geodetic measurement methods are also covered as they are often used in conjunction with geotechnical and structural instrumentation. These methods include total stations, levelling and differential RTK GPS, and more recent methods such as LiDAR (Light Detecting and Ranging), ground-based InSAR (Interferometric Synthetic Aperture Radar), and satellite InSAR, and photogrammetric techniques using air-borne drones.

25.2 Geotechnical Instrumentation and Monitoring Information Sources

The most recognized source of **geotechnical instrumentation and monitoring** information is the 575-page book written in 1988 and re-edited in 1993 by John Dunncliff: *Geotechnical Instrumentation for Monitoring Field Performance* (Dunncliff 1993), colloquially called the “Red Book”. John Dunncliff has also authored ninety-three quarterly columns on geotechnical instrumentation in the former *Geotechnical News* magazine of the Canadian Geotechnical Society (CGS). An index and the searchable content of articles that were published as part of these columns is archived in the public section of the CGS website at (https://www.cgs.ca/instrumentation_news.html).

Comprehensive information on Geotechnical Instrumentation has also been published in the symposia “Field Measurements in Geomechanics”: 1983, Zurich, Switzerland; 1987, Kobe, Japan; 1991, Oslo, Norway; 1995, Bergamo, Italy; 1999, Singapore; 2003, Oslo, Norway; 2007, Boston, USA; 2011, Berlin, Germany; 2015, Sydney, Australia and 2018, Rio de Janeiro, Brazil. These were organized by a group of volunteers, mostly academics, and the proceedings have been published by commercial publishers. The next symposium will be in London in 2022 and will be sponsored by the International Society for Soil Mechanics and Geotechnical Engineering (ISSMGE).

The Institution of Civil Engineers (ICE) Manual of Geotechnical Engineering (Burland et al, 2012) contains two chapters of interest on Geotechnical Instrumentation: Chapter 94: Principles of Geotechnical Monitoring (16 p.) and Chapter 95: Types of Geotechnical Instrumentation and Their Usage (26 p.).

ICE has also produced another publication of interest: Client Guide to Instrumentation and Monitoring, by the Chartered Institution of Civil Engineering Surveyors for the Survey Liaison Group, London, 2017 (28 p.).

The American Society for Testing and Materials (ASTM) and International Organization for Standardization (ISO) have issued standards that cover some of the geotechnical instrumentation applications.

The ASTM Subcommittee D18.23 on Field Instrumentation has published the following standards:
D4403-20 Standard Practice for Extensometers Used in Rock

D6230-13 Standard Test Method for Monitoring Ground Movement Using Probe-Type Inclinometers

D6598-19 Standard Guide for Installing and Operating Settlement Points for Monitoring Vertical Deformations

D7299-12 Standard Practice for Verifying Performance of a Vertical Inclinometer Probe

D7764-12 Standard Practice for Pre-Installation Acceptance Testing of Vibrating Wire Piezometers

An effort has also been initiated by the ISO Technical Committee 182/WG2 on Monitoring in Geotechnical Engineering to produce standards on geotechnical monitoring by field instrumentation. Five standards have been published to date:

ISO 18674-1:2015 Geotechnical investigation and testing — Geotechnical monitoring by field instrumentation. Part 1: General rules.

ISO 18674-2:2016 Geotechnical investigation and testing — Geotechnical monitoring by field instrumentation. Part 2: Measurements along a line - extensometers.

ISO 18674-3:2017 Geotechnical investigation and testing — Geotechnical monitoring by field instrumentation. Part 3: Measurements across a line - inclinometers.

ISO 18674-4 Geotechnical investigation and testing — Geotechnical monitoring by field instrumentation. Part 4: Measurement of pore water pressure: piezometers.

ISO 19674-5 Geotechnical investigation and testing — Geotechnical monitoring by field instrumentation. Part 5: Stress change measurements by total pressure cells.

Additional ISO standards that are currently planned to be published in the coming years include: Part 6 Hydraulic Settlement Gauges, Part 7 Strain Gauges, Part 8 Load Cells, and Part 9 Geodetic Monitoring Instruments.

The Transportation Research Board (TRB) published a comprehensive technical note on the “Use of Inclinometers for Geotechnical Instrumentation on Transportation Projects” (Machan and Bennett, 2008).

Published textbooks include:

De Rubertis, K. (Editor), 2018. Monitoring Dam Performance-Instrumentation and Measurements, Task Committee to Revise Guidelines for Dam Instrumentation of the Committee on Water Power of the Energy Division, American Society of Civil Engineers (442 p).

ICOLD (International Commission on Large Dams) Bulletin 158, Dam Surveillance Guide (2018) (109 p).

Sharon, R., Eberhardt, E. (Editors), 2020. Guidelines for Slope Performance Monitoring, CSIRO Publishing, CRC Press/Balkema (331 p).

Stark, T.D., Oommen, T., Zhangwei Ning, Z. 2021. Remote Sensing for Monitoring Embankments, Dams, and Slopes: Recent Advances, ASCE GSP 322. Sponsored by the Embankments, Dams, and Slopes Technical Committee of the Geo-Institute of ASCE (114 p).

Published Instrumentation Guidelines related to dam safety monitoring and which are available for download include:

U.S. Army Corps of Engineers (2020). Instrumentation of Embankment Dams and Levees, Engineer Manual EM-1110-2-1908, 30 November (290 p).

U.S. Army Corps of Engineers (1987). Instrumentation for Concrete Structures, Engineer Manual EM-1110-2-4300, 30 November (306 p). Currently under revision.

U.S. Dept. of Interior, Bureau of Reclamation (Reclamation) (1987). Embankment Dam Instrumentation Manual. Bartholomew, C.L., Murray B.C., Goins, D.L., U.S. Department of the Interior, January (269 p). Currently under revision.

U.S. Dept. of Interior, Bureau of Reclamation (Reclamation) (1987). Concrete Dam Instrumentation Manual. Bartholomew, C.L., Haverland, M.L., U.S. Department of the Interior, October (153 p).

FERC (Federal Energy Regulatory Commission) Engineering Guidelines for the Evaluation of Hydropower Projects (2005), Chapter 9, Instrumentation and Monitoring (86 p).; Chapter 14 (2017), Dam Safety Performance Monitoring Program (188 p).

U.S. Society on Dams (2008-2020), Series of 7 White Papers on the theme of Development and Implementation of Dam Safety Monitoring Programs.

Comisión Nacional del Agua, Mexico (2012). Manual de Mecánica de Suelos: Instrumentación y Monitoreo del Comportamiento de Obras Hidráulicas (322 p).

25.3 Human Factors in Geotechnical Instrumentation and Monitoring

Dunncliff (1993) devotes three chapters to topics related to human factors in geotechnical instrumentation and monitoring. These include Systematic Approach to Planning Monitoring Programs Using Geotechnical Instrumentation, Specifications for Procurement of Instruments, and Contractual Arrangements for Field Instrumentation Services. Although written several years ago, the opinions and systematic approach steps suggested by this author remain valid today, if not even more so, due to the faster pace of today's construction projects.

The historical context provided by Dunnicliff (1993) in his introductory Chapter 1 illustrates why he has always given so much emphasis during his career to Human factors: *“The birth of geotechnical instrumentation, as a tool to assist with field observations, occurred in the 1930s and 1940s. During the first 50 years of its life, a general trend can be observed. In the early years, simple mechanical and hydraulic instruments predominated, and most instrumentation programs were in the hands of diligent engineers who had a clear sense of purpose and the motivation to make the programs succeed. There were successes and failures, but the marriage between instruments and people was generally sound. In more recent years, as technology advanced and the role of geotechnical instrumentation became more secure, more complex devices with electrical and pneumatic transducers have become commonplace. Some of these devices have performed well, while others have not. At the same time, the technology has attracted an increasingly large proportion of the geotechnical engineering profession, and an increasing number of instrumentation programs have been in the hands of people with incomplete motivation and sense of purpose. There have continued to be successes and failures but, in contrast to the early years, a significant number of the failures can be attributed to an unsound marriage between instruments and people.”*

Further views on similar topics of Contract Practices for Geotechnical Instrumentation are also provided in Dunnicliff (1994) and in Chapter 94 of Burland et al. (2012).

25.4 Measuring Principles Commonly Used in Geotechnical Instrumentation

Geotechnical instruments are based on electrical, mechanical, hydraulic, and pneumatic measuring principles. The two main electrical measuring principles are vibrating wire and Micro Electro-Mechanical Systems (MEMS) accelerometers circuits and are described in the following sub-sections. Other electrical measuring principles which are more prevalent in industrial instrumentation and process control such as piezoresistive circuits, potentiometers and Linear Variable Differential Transformers (LVDT) are also used occasionally in geotechnical instruments (although not discussed herein). Mechanical (e.g., a dial depth gage or a micrometer) and hydraulic (e.g., hydraulic piezometer which was quite prevalent in embankment dams in the ‘60s and ‘70s or a manometer for settlement measurement) measuring principles are also not discussed herein, since they have largely been replaced by vibrating wire and MEMS instruments. Pneumatic piezometers are still used, especially for short term applications such as preloads on soft soils, although to a much lesser extent than their vibrating wire counterparts. In some cases, users consider pneumatic piezometers as a redundant type of piezometer in addition to vibrating wire ones.

25.4.1 Vibrating Wire Instruments

The vibrating wire principle is by far the most prevalent for electrical instruments in geotechnical instrumentation. Vibrating wire technology has been known since early in the 1900s and was initially used mostly in concrete dams in Europe after WW2, where it earned its reputation for robustness and long-term stability and durability. Early vibrating wire instruments were mostly strain gauges and pressure cells, while piezometers became available in the 1950’s when they were initially installed in embankment dams. Today, vibrating wire instruments encompass the full range of parameters to be measured in geotechnical instrumentation, including pore and interstitial water pressure, settlement, extension/compression, strain, load, stress, and earth pressure. Temperature measurement is often incorporated in vibrating wire instruments using a hermetically sealed thermistor.

The working principle of the vibrating wire is a simple one, based on a small diameter (typically 0.25 mm) steel wire which is pretensioned between two end points. The length of the steel wire can be 5 to 15 cm depending on the type of instrument. The measuring principle is based on the resonant oscillating frequency of the wire when it is artificially excited by a readout unit or datalogger using two small electromagnetic

coils mounted along the length of the wire. One of the coils is magnetized by a small voltage frequency sweep that induces the vibration of the vibrating wire at its resonant frequency, while the other coil generates an AC signal whose frequency is the same as the resonant frequency of the wire, typically in the 1000 to 2000 Hz range. Since the resonant frequency varies with the tension in the wire, sensors to measure pressure, strain and elongation can be built by mounting the vibrating wire in the sensor and calibrating the resonant frequency as a function of the parameter to be measured. More details on the vibrating wire principle and its suitability for geotechnical instrumentation applications can be found in McRae et al. (1991) and Choquet et al. (1999).

25.4.2 Micro-electromechanical Systems (MEMS)-Based Instruments Digital Output Instruments

Parameters typically not measured using the vibrating wire principle include tilt and inclination, although vibrating wire tiltmeters are still available from some manufacturers. Tilt and inclination have traditionally been measured with servo-accelerometers and electrolytic tilt sensors until about 2010 where they were superseded by MEMS accelerometers which have excellent metrological properties, at least equal to their predecessors, with the added advantage of better robustness and temperature offset properties (Sellers et al., 2008). Electronic circuits that incorporate MEMS accelerometers can include a micro-controller and analog to digital converter so that the resulting tilt and inclination instruments can have a digital output and be bussed on a single 4-conductor cable using an RS-485 protocol or another similar digital protocol. MEMS-based instruments often incorporate a solid state temperature sensor on the circuit board.

25.4.3 Digital Output Instruments

In addition to MEMS-based tilt and inclination instruments, other types of instruments can have a digital output. For example, piezoresistive water level sensors with a 4-20mA output are commonly used in groundwater monitoring, but they can also be used as piezometer when a porous filter similar to the one used in vibrating wire piezometers is positioned in front of the diaphragm. These piezoresistive instruments can incorporate an analog to digital converter and be read on a RS-485 digital bus.

Other types of digital instruments are commonly used in groundwater monitoring, including multi-parameter probes for water quality, conductivity, and pH. Here, the digital communication protocol can be RS-485, but another prevalent digital protocol is SDI-12 that was developed several years ago by the U.S. Geological Survey. Many dataloggers and portable readout units can read digital instruments that use those protocols.

Digital thermistor strings have been available commercially for many years. They are described in Section 25-14. Although not yet available commercially, a next step would be strings of piezometers that would incorporate a vibrating wire to digital converter to make digital piezometer strings that would be connected on a single 4-conductor cable.

25.4.4 Fiber-Optic Instruments

Another measuring principle gradually becoming more prevalent is fiber optic sensing. Fiber optic sensors are immune to lightning and high electrical transients which is a significant advantage in many applications. Point sensors like piezometers, displacement transducers, and strain gauges are available in fiber optic versions, but the main difference is that fiber optic sensors are also available as semi-distributed or fully distributed sensors. Semi-distributed sensors, most of them based on Bragg gratings, involve several measuring points on a single fiber optic cable to measure strain and temperature. Fully distributed sensors based on Brillouin scattering allow strain and temperature measurements every metre over fiber optic cables that can be as long as 30 km. Fiber optic sensors are available from several commercial sources and are the

subject of considerable research to develop even more field applications. Glisic and Inaudi (2007) provide an excellent reference for a summary of principles and applications of fiber optic sensing.

25.5 Metrological Properties of Geotechnical Instruments

Due to the context in which they are used, geotechnical instruments are expected to be robust, accurate and have excellent longevity. In many cases, geotechnical instruments cannot be easily replaced if they fail, or the cost of replacing them would be very high. For this reason, excellent metrological properties are expected from them. The main metrological properties are as follows (De Rubertis, 2018):

- Resolution is the smallest change in the measured parameter that can be distinguished by an instrument and is typically many times finer than the instrument's precision, repeatability, and accuracy.
- Accuracy is the degree to which readings match an acceptable standard. Most geotechnical instruments from North American manufacturers are calibrated using standards traceable to the National Institute of Standards and Technology (NIST). It is most commonly written as a plus-minus (+/-) percentage from the full-scale range (F.S.), such as +/- 0.1% F.S. Vibrating wire instruments typically have an accuracy of +/- 0.1% F.S., a similar value to what is encountered for the other measuring principles.
- Precision, which could also be called random error, is the closeness of similar measurements to the arithmetic mean. It is expressed as a plus-minus (+/-) value. In geotechnical monitoring applications, precision is quite often more important than accuracy because change, rather than absolute value is of greatest interest. Vibrating wire instruments typically have a precision of +/- 0.025 % F.S., also very similar to the other measuring principles.
- Repeatability is how closely several consecutive measurements agree with one another under the same operating conditions. It is expressed in engineering units such as 0.1 mm or as a percentage of full-scale range. In practice, repeatability is the same as precision for vibrating wire instruments.
- Linearity is the closeness to which a curve of instrument output versus input approximates a straight line. It is usually measured as nonlinearity, e.g., the maximum deviation between the curve and a straight line and expressed as a percentage of the full span of output of the instrument. Vibrating wire instruments typically have a non-linearity of 0.1 to 0.5 % F.S. To improve the non-linearity properties, the instrument output can also be fitted to a second or higher order polynomial equation.
- Bias is the average indicated value of the measured variable that would result from an infinite number of the measurements minus the actual value of the variable. In practice, bias is the same as the accuracy of the instrument.
- Long-term stability is an indication of the change in an instrument's output that would take place over a specified period of years in the absence of a change in the measured input (pressure, displacement, tilt, etc.) Because of the long time periods involved in gathering data and the diversity of factors that can contribute to output drift over these periods, precise answers can be elusive. However, case histories reporting the long-term performance over tens of years of geotechnical instruments, especially of the vibrating wire type, are often reported in literature, including in McRae et al. (1991) and Choquet et al. (1999).

25.6- Static vs Dynamic Monitoring

This chapter's main focus is on static measurements or measurements collected a few times per day as opposed to dynamic measurements which are performed several times per second during transient

phenomenon such as seismic events or transient loading such as vehicle passage over a bridge or a soft embankment. Situations where dynamic measurements are also technically feasible will however be mentioned.

Static measurements are generally preferred for long-term performance monitoring of structures since measurements are collected and stored using the same monitoring equipment and the relatively small quantity of data generated is easier to visualize. However, current technology allows continuous dynamic measurements to be taken which can then be buffered and automatically deleted in the absence of a transient phenomenon.

There are technologies which parallel the field of instrumentation and monitoring that make use of dynamic measurements. Some of them are used in site investigation, for example, CPT (Cone Penetration Testing) and some geophysical methods like seismic reflection and seismic refraction surveys. PDA (Pile Dynamic Analysis) testing is another example. However, these methods are not discussed herein.

25.7- Sensors, Transducers, Transmitters, and Instruments

In the context of geotechnical instrumentation, the words sensor, transducer, transmitter, and instrument tend to be used almost interchangeably. However, in principle, a sensor is a device that senses a physical quantity and converts it into an analogue quantity which can be measured in some way, for example electrically as a voltage, capacitance, inductance or ohmic resistance. A transducer is also a device that measures a physical quantity, but it also converts one form of energy into another. In other words, a transducer is a sensor with some sort of signal conditioning to provide a more readable signal. As an example, a piezoresistive pressure sensor which is based on a strain gauge principle, usually generates a low-level voltage signal. If that signal is converted into a higher voltage signal, for example 0-5 Volts, using signal conditioning in the device, it becomes a pressure transducer. The word transmitter comes from the process control industry where a transducer that converts into a standardized instrumentation signal such as a 4-20 mA signal is called a transmitter. And in all three cases, they will be read by an instrument which displays the measured voltage or the converted engineering unit such as kPa.

By definition, a vibrating wire piezometer is not an instrument, but rather a transducer that incorporates a sensor (the diaphragm/vibrating wire assembly) that converts the pressure into a frequency signal. It is read using an instrument which can be a portable readout or a datalogger. The word instrument will however still be used herein, as it has now become common language in geotechnical engineering.

25.8 Measurement of Groundwater Pressures

Groundwater pressures are measured with piezometers. A piezometer is a device installed at depth sealed in the ground in such a way that it will measure the water pressure representative of the depth at which it is installed. Piezometers can consist of simple Casagrande standpipe piezometers or specialized electric pressure transducers installed at depth in boreholes. Hydraulic and pneumatic piezometers are also an option, especially pneumatic types. Their working principle and installation procedures are described in Dunnycliff (1993).

25.8.1 Casagrande Standpipe Piezometers

A simple and very common piezometer is an open standpipe, often called a Casagrande standpipe piezometer. It is made of two parts: a porous filter or slotted well screen and a standpipe consisting of flush-threaded PVC pipes. The annulus around the porous filter or well screen is backfilled with clean, well draining sand, and a bentonite seal is created above and often below the sand zone. The rest of the borehole

is usually backfilled to the surface with a cement-bentonite (sealing) grout or sometimes drill cuttings. The water level in the standpipe is measured with a water level meter which is a probe mounted on a graduated tape lowered in the borehole and which establishes an electrical circuit which sounds a buzzer when in contact with water.

Although not easy to achieve, a second standpipe can be created above the first one by lowering a second porous filter or wells screen and PVC standpipe and repeating the sand backfilling and sealing process. However, when increasing the number of standpipes in the same hole, the installation of filter sand and bentonite becomes increasingly difficult.

When groundwater sampling for water quality is also required, for example when the objective of the installation is both geotechnical and environmental, a 50 mm Schedule 40 PVC or stainless-steel pipe with flush-threaded pipes and screened bottom section is often preferred. Couplings for PVC pipes should not make use of solvent (cement) if water quality sampling is required. Rather, pipe connections should be made with a thread/sealing system, conforming to ASTM F480, using precision threads, an “O”-ring seal and a wedge locking mechanical seal to obtain a strong, water-tight connection.

It should be kept in mind that the larger the diameter of the standpipe, the longer it will take for the water level to reach static equilibrium. In fine grained soils, the time lag can be several months. If shorter times are desired to reach equilibrium, consideration should then be given to using vibrating wire piezometers.

The diameter of the standpipe should also consider the ease of decommissioning of the well. When installing a piezometer with a diameter of 32 mm or greater, it is possible to install a tremie tube to grout the tube when decommissioning the well. With smaller diameter piezometers, the tremie tube is difficult or impossible to fit in the pipe and a drill rig may be needed to over-drill and properly backfill the installation.

A common, although possibly flawed practice in some circumstances, is to install a 50 mm combined piezometer and environmental well while using a screen with the finest (# 10) slots and a coarse sand (# 2, #3 or #4). While the use of coarse sand reduces installation time (the sand drops into the drill hole faster if it is coarse grained, in particular through water) it may be too coarse to filter out fine sands and silt. Upon preservation of the sample (with acid), metals in the soil may be leached out with resulting impacts on water quality test results.

There are also commercially available multi-level observation well systems that allow multiple piezometers to be put into a single borehole while allowing sampling and pore pressure measurement at discrete intervals (Westbay system, Waterloo system).

25.8.2 Electric Pressure Transducers

Electric pressure transducers specifically designed for water pressure measurement in soil and rock are called piezometers. The most common are based on the vibrating wire principle and are generally referred to as Vibrating Wire (VW) piezometers (Figure 25-1).

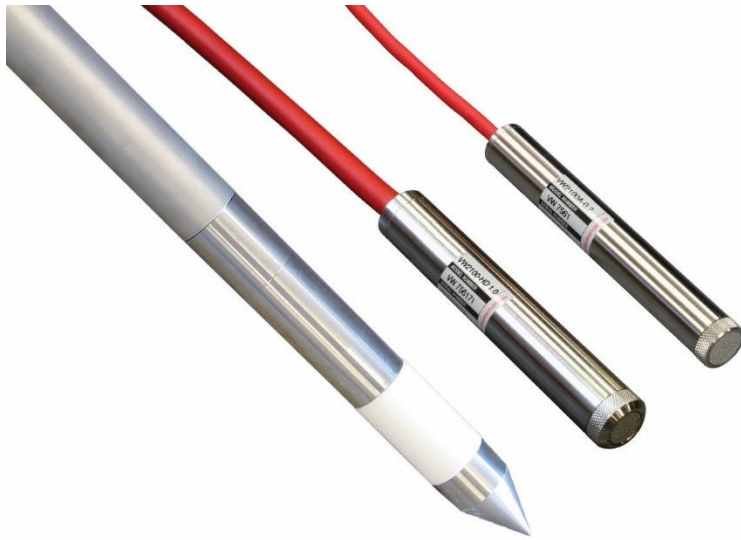


Figure 25-1 VW Piezometers. Left to right: drive point (push-in), heavy duty and standard

Piezometers based on piezoresistive circuits (also called strain gauge piezometers) are also available, although more common in the groundwater monitoring industry. In the case of groundwater monitoring, they are usually installed in cased boreholes to measure the water level or phreatic surface.

Piezometers used in geotechnical instrumentation have a porous filter tip to prevent the diaphragm of the measuring insert from being in direct contact with the soil or the grout in which the piezometer is installed, and therefore measure true pore pressure and not total pressure. Two types of porous tip filters are available: Low Air Entry (LAE) and High Air Entry (HAE). The air entry pressure refers to the air or gas pressure required to extrude the water contained in the filter once it is pre-saturated. In the majority of cases, LAE filters should be used. HAE filters have been advocated by some authors because of their theoretical capability of allowing measurement of negative pore pressure in unsaturated soils. However, it is currently believed by some practitioners (Penmann, 2002, Sellers et al., 2003) that piezometers with HAE filters may lose their saturation with time because of cavitation phenomena and that they then lose their ability to measure positive pore pressure.

VW piezometers can be installed in a Casagrande standpipe piezometer, inside a sand zone at the bottom of a borehole which is subsequently backfilled, pushed in place using a drill rig or fully-grouted in a borehole with a bentonite-cement mix. The four methods are discussed in the following sections.

25.8.3 VW Piezometer in a Casagrande Standpipe

The VW piezometer can be suspended from the collar of the borehole to a predetermined depth below the anticipated water level in the standpipe. It is a good practice to read the water level with a portable water level meter before lowering the VW piezometer for comparison. It should be noted that the volume of the cable and of the VW piezometer will change the water level temporarily in the borehole until stabilization within a few hours or days, depending on the permeability of the surrounding soil. It is good practice to consider using a Kevlar reinforced signal cable if the water level is more than 30 to 50 m below ground surface, as a non reinforced cable will stretch with time.

25.8.4 Zoned Installation in a Borehole

The VW piezometer is pre-inserted into a sandbag, then lowered to the bottom of the borehole, typically an HWT-cased borehole (122.6 mm diameter), or a hole drilled with either 150 mm hollow stem augers or 125

mm solid stem augers. Then, as the casing or auger retracted, sand is added to create a filter zone approximately 1 metre long around the piezometer. Bentonite chips are then added to create a seal and define the zone of pore pressure measurement. It is a good practice to add water to just above the seal zone first and install the sand and bentonite in water in order to reduce bridging. The remainder of the borehole can be backfilled with a cement-bentonite grout or with a sealing grout. A second or possibly third measurement zone can be created by repeating this procedure, although it may become increasingly difficult to create proper seal between each zone.

25.8.5 Push-In Piezometers

Push-in piezometers (also called drive-point piezometers) are pushed in place, most often using the drill rig head. They are generally used in soft clays or similarly soft materials. Their installation depth is generally limited to about 30 metres if installed from the surface, or they can be installed deeper by drilling a pilot hole and pushing the piezometer an additional metre from the bottom. The porous filter of the push-in piezometer is best positioned along its side (as opposed to the tip location in the case of standard piezometers) to be in direct contact with the soil (Figure 25-1). Excess pore pressures will typically develop during the pushing process and care should be taken to avoid exceeding the instrument's measurement range. The excess pore pressure will usually stabilize within a few hours to one day. Saturation of the filter in a vacuum using glycerine prior to installation reduces the potential to lose saturation during pushing and may also shorten the stabilization period.

25.8.6 Fully-Grouted Piezometers

Grouting a VW piezometer in a continuous column of cement-bentonite grout was introduced by Vaughan (1969) and more extensively by McKenna (1995). It became mainstream after the publication of three landmark papers (Mikkelsen and Green 2003; Contreras et al. 2008; Contreras et al. 2011) that explained in very clear terms, the theoretical concepts, and the reasons for which this installation method is suitable in many if not most situations. Practitioners are advised to read these three references to understand the principles, proper installation techniques, and limitations of use for this installation method. The installation process of fully-grouted piezometer is quite simple once the cement-bentonite grout mix composition and mixing method have been defined.

The main concept of the fully grouted method is that the cement-bentonite grout should have a permeability and a strength similar to the that of the surrounding soil formation in which the VW piezometer is installed. Contreras et al. (2008) argued that the permeability of the grout can be up to three orders of magnitude higher than the permeability of the formation and porewater pressures will still not be affected by upward or downward groundwater flow in the grout column. For example, in a clay formation of say 10^{-9} cm/sec, the grout could have a permeability between 10^{-9} cm/sec and 10^{-6} cm/sec. If, however, the permeability of the grout is lower than that of the formation, then a longer response time should be expected.

Mikkelsen and Green (2003) presented two grout mixes that have been adopted by practitioners and have shown good performance (Table 25-1). In rock formations, most practitioners use the grout mix for medium to hard soils. These grout mixes are the same as those recommended for inclinometer casing.

Application	Grout for Medium to Hard Soils		Grout for Soft Soils	
Materials	Weight	Ratio by Weight	Weight	Ratio by Weight
Water	30 gallons (113.5 liters)	2.5	75 gallons (284 liters)	6.6
Portland Cement	94 lbs (1 sack) (42.6 kg)	1	94 lbs (1 sack) (42.6 kg)	1
Bentonite	25 lbs. (as required) (11.3 kg)	0.3	39 lbs. (as required) (17.7 kg)	0.4
Notes	The 28-day compressive strength of this mix is about 50 psi (340 kPa), similar to very stiff to hard clay. The modulus is about 10,000 psi (69 Mpa).		The 28-day strength of this mix is about 4 psi (27 kPa), similar to very soft clay.	

Table 25-1 Cement-bentonite grout, water and cement mixed first (after Mikkelsen and Green 2003).

The authors also add the following recommendations in their paper:

“In order to keep field procedures simple, the emphasis should be on controlling the water-cement ratio. This is accomplished by mixing the cement with the water first. When water and cement are mixed first, the water-cement ratio stays fixed and the strength/modulus of the set grout is more predictable. If bentonite slurry is mixed first, the water-cement ratio cannot be controlled because the addition of cement must stop when the slurry thickens to a consistency that is still pumpable.

Making cement-bentonite grout in the field is a straightforward process. The most effective mixing is commonly done in a barrel or tub with the drill-rig pump, circulating the batch through the pump in 50 to 200-gallon quantities. The rig pump provides the kind of jet-mixing required to get the job done quickly. Any kind of bentonite powder used to make drilling mud combined with Type 1 or 2 Portland cement and water can be used, but the appropriate quantity of bentonite will vary somewhat depending on grade of bentonite, mixing sequence, mixing effort (agitation), water pH and temperature. Grout mixes should be controlled by weight and proportioned to give the desired strength of the set grout.”

Paddle or high shear mixers can also be used to mix the grout. Other useful and more detailed recommendations for preparing the cement-bentonite grout are presented in Mikkelsen (2002).

It is important to mention that the fully grouted piezometer installation method is not accepted by all practitioners with some preferring the zoned installation described above for the following reasons/concerns (also discussed in the three referenced landmark publications mentioned above):

- Zoned installation better defines a measurement zone,
- Mixing of the grout may not be done properly,
- Grout permeability may be far different than planned,
- The cement or bentonite used may be incompatible with the chemical composition of the groundwater,
- The grout might crack if subject to excessive vertical settlement, or may bleed and dissipate in a high permeability formation,
- In mud rotary drilling with casing, smearing with fine cuttings may form along the borehole wall and may create an annulus of low permeability material causing increased vertical groundwater flow,

- An adjacent formation with high permeability and pressure gradient may cause erroneous readings of a grouted piezometer installed in a less permeable formation, especially when the grouted borehole crosses both formations, and
- Installation in artesian flow conditions can be difficult.

The above concerns illustrate that piezometer installation must be planned, properly specified and that the working principles and theory of fully grouted piezometers must be well understood. More recent technical papers highlight that the use of fully-grouted piezometers may be less satisfactory in conditions where transient seepage conditions are present, as discussed by Martens et al. (2020) and Marefat (2018), and that the allowance suggested by Contreras et al. (2008) where the permeability of the grout can be up to three orders of magnitude higher than the permeability of the formation is too permissive and should be reduced to one order of magnitude or less. Martens et al. (2020) provided the following recommendations based on the conditions they have modelled for the piezometer readings of transient pore pressure in clays, in the situation of a sand embankment constructed rapidly over a low permeability clay formation and a fully-grouted piezometer installed through the two formations:

- Errors for readings under steady state flow conditions may be acceptable with a grout/formation permeability ratio of 100. Under these conditions, with a constant gradient across the clay layer, the error is greatest at greater distance from the sand-clay interface,
- For transient flow, the errors are small for a 10 times permeability ratio between the grout and formation, at all practical distances from the interface, and all times beyond the first few days following loading,
- At a 100 times permeability ratio, the errors can be large near a stratigraphic interface and for times in the range of weeks to months. After a year or more, the error becomes small.
- At a 1000 permeability ratio, the errors can be large for greater distances from the interface, and for extended periods of time, and
- For conditions where the pore pressure increase uniformly in a clay layer due to application of an external load where there is no initial gradient across the clay layer, the error is greatest near the sand-clay interface.

The use of a grout mix with a permeability in the order of 1×10^{-7} cm/sec has recently been suggested by Contreras et al. (2020) recognizing that in some jurisdictions, geotechnical boreholes may be considered the same as environmental boreholes. In such cases, legislation may impose stringent requirements on the permeability of the grout mix to be used in fully-grouted piezometer installations to avoid the communication of groundwater between aquifers. Some regulations require the grout used to fill environmental boreholes must have a permeability ranging from 1×10^{-7} to 1×10^{-8} cm/s or lower. To meet these more stringent requirements, a grout mix with a water:cement:bentonite ratio (by weight) of 2.0:1.0:0.36 with 140 g of a water reducer super-plasticizing additive (e.g., Specrete-Flow Aid HR) per 43 kg bag of cement is proposed. The additive achieves a Marsh Funnel viscosity more compatible with pumps commonly found on geotechnical drill rigs. Contreras et al. (2020) report that this new mix “shows a Marsh Funnel viscosity of about 60 seconds. This indicates that the additive is very effective in providing an acceptable Marsh Funnel viscosity so that the grout is easily pumpable. The UCS is 1,216 kPa and the permeability is 2.4×10^{-7} cm/s, e.g. within the acceptable range required by the regulators.” This new grout mix with lower permeability is therefore a new mix that is available in the portfolio of mixes, especially when a grout with low permeability is required for reasons of permeability compatibility with the enclosing ground.

Most installations in soil will be made using either HWT-cased mud rotary, hollow or solid stem augers, or sonic drilling. Once the borehole reaches its target depth, one or more VW piezometers are fixed (often with electrical tape) to a 20 to 25 mm diameter PVC tremie pipe perforated at its end to allow better grout outflow. A spacer can be positioned between the body of the piezometer and the tremie pipe so that the

piezometer will be surrounded by grout and minimize the possibility of future vertical hydraulic conductivity along the outside of the pipe. For the same reason, the signal cables should not be taped tight along the tremie, every 2 or 3 m probably being sufficient, and they should not be bundled together. The tremie pipe is then lowered down the borehole and the grout pumping can proceed at the same time as the casing, or the auger is pulled out of the borehole (or in an open hole).

Care must be taken in selecting the measuring range of the VW piezometer so that the weight of the grout column, which usually has a specific gravity of around 1.3, does not exceed the range of the piezometer. It is not necessarily good practice to rely upon an allowable temporary overrange of 50 or 100 % as mentioned in the manufacturer documentation, but rather, a higher range device should be used. In the case of deep installations, a staged grouting with 2 or more tremie pipes at different depths can also be considered, especially for deep installations in rock.

It is good practice to take frequent readings during and immediately after grouting and for the next 24 to 48 hours to confirm stabilization of the readings while the grout is curing. Generally, readings will be almost fully stabilized after about 24 hours. The curing temperature of the grout will also be measurable with the incorporated thermistor in the VW piezometer. It is not unusual to read temperatures up to 50° C immediately after grouting, and to see the temperature stabilize to the natural ground temperature, often in the 5 to 15° C range within 24 hours.

A standard vibrating wire piezometer is hermetically sealed and therefore can also respond to changes in barometric (atmospheric) pressure. Measuring barometric pressure is optional as some users choose to not do a barometric correction of their piezometer readings. Barometric correction will correspond to about +/- 25 cm of water head (+/- 25 kPa) change between a high-pressure day and a low-pressure day, so the user can decide if that value is of importance in case of short- or long-term monitoring programs. Another consideration on barometric correction is that piezometers installed at greater depths or sealed in a borehole may not be influenced by barometric pressure changes. In such cases, a barometric correction would introduce an error equivalent to the correction made. To assess if piezometers are influenced by changes in barometric pressure, uncorrected piezometer measurements and barometric pressure measurements can be plotted on the same graph. If the piezometer measurements follow the same trend as the barometric pressure, variations in barometric pressure are being measured and a correction may be appropriate.

Vented VW piezometers can be used to avoid the need for barometric corrections. However, for long term applications, unvented VW piezometers are preferred as the vent tube of vented piezometer requires a desiccant cartridge that must periodically be changed when it becomes saturated. Often this change is neglected, and humidity can ingress the vent tube, forming a water meniscus which will prevent proper venting and cause erratic readings.

Finally, last but not least, as these things happen, don't forget to take the initial reading of the VW piezometer at the surface prior to lowering it in the borehole, as this initial reading will be the baseline from which water pressure will be calculated. The piezometer should be saturated prior to taking the initial reading following manufacturer's recommendations. Good practice is to remove the filter of the VW piezometer and immerse the filter and the piezometer in water for at least 30 minutes, then reattach the filter. The water temperature should be stable, e.g., the water bucket should not be directly in the sun. Agitation of the water will help remove air bubbles that may be stuck to the walls of filter. Then the piezometer is lifted from the bucket just above water and the initial reading, the temperature from the thermistor, and the barometric pressure are recorded.

25.8.7 Measurement of Negative Pore Pressure (Suction)

Suction, or negative pore pressure, is generally encountered in the vadose zone, which is the unsaturated zone of soil formations, typically near the surface. Negative pore pressures generally aid slope stability and a design that does not consider them may be on the conservative side if groundwater conditions remain relatively consistent. It can be of interest to measure negative pore pressures for various reasons, including understanding the seasonal saturation and desaturation of the vadose zone.

Ideally, a method that would measure negative pore pressure would be able to measure both negative and positive pressures and remain operational when pore pressures alternate between negative and positive. The element that complicates the measurement of negative pore pressure is air and/or gas which is often present in pores in unsaturated soils. De-aired water can transmit tensions and therefore an electric piezometer diaphragm which is in contact with de-aired water under tension will be able to measure negative pressures. The difficulty in measuring negative pore pressure in practice is ensuring that the water in contact with the diaphragm remains de-aired. This can be done using a ceramic filter with a high air entry pressure. Air entry pressure of a filter, also called bubbling pressure, is defined as the pressure differential at which blow-through of gas occurs. Thus, a filter with a high air entry value (or high bubbling pressure) will allow a high pressure differential before blow-through occurs.

High air entry filters are made of a porous ceramic with a pore diameter of about 1 micron, providing an air entry value of at least 100 kPa and a permeability to water of about $3 \cdot 10^{-6}$ cm/sec. This means that high air entry filters can theoretically measure negative pressures up to the air entry value. As a comparison, low air entry filters that are used with standard VW piezometers in saturated soils are coarse filters that readily allow passage of both gas and water. Typical low air entry filters are made of sintered stainless steel and have a pore diameter of 20-80 microns, providing an air entry value in the of 3 to 30 kPa. A filter with a pore diameter of 50 microns has a permeability to water of about 3×10^{-2} cm/sec. (Dunnicliff, 1993).

Methods for measuring negative pore pressures include the following:

- A VW piezometer with a high air entry filter saturated in the factory with de-aired water and shipped to site in a bottle filled with de-aired water. On site, prior to installation, the cavity behind the filter and the diaphragm should be filled with de-aired water. This method applies to piezometers that would then be inserted directly in the fined grained material in a small cavity of the same shape as the piezometer. Some push-in piezometers can also have high air entry filters and can therefore be used in the same way with the same care in the saturation process.
- Using a twin-tube hydraulic piezometer which is a piezometer design with a high air entry ceramic filter that was used extensively in the 1950s and 1960s but is almost not used anymore as it requires periodic flushing of the fluid with de-aired water (Dunnicliff, 1993). However, this requirement of periodic flushing is favourable for measuring negative pore pressures as it can restore the capability of reading negative pressures if air or gas has gradually percolated through the ceramic filter.
- Installing a VW piezometer in a fully grouted borehole. Mikkelsen and Green (2003) argue that cement-bentonite grout offers advantages for pore water suction measurements in partially saturated soils. They point out that *“cement-based grouts have an air entry value up to about 100 psi, a property required for measurement of negative pore pressures. The grout, when set, becomes a perfectly saturated porous material, a far easier task than pre-saturating a high-air entry ceramic filter and deploying it down a borehole. A vibrating wire piezometer fully grouted within cement-bentonite grout can therefore be expected to measure negative pore water pressure or suction”*.

It should be noted that some users have anecdotally reported observing negative pore pressure measurements with fully-grouted VW piezometers, but this subject has not yet been fully validated by published research.

- Using a flushable VW piezometer that has been especially developed for the purpose of measuring both positive and negative pore pressure, which incorporates a flushing mechanism to periodically flush the fluid behind the filter and remove gas bubbles that may have percolated through the ceramic filter (Ridley, 2015). This piezometer needs also to be grouted in a borehole in the same way as fully-grouted piezometers.

A general recommendation about measuring negative pore pressure would be that the methods described above may only properly operate a short time until gas percolates behind the filter and compromises the measurements. Using a flushable VW piezometer may be more reliable though, having been specifically developed for that purpose. All methods should be used with care and a good understanding of their operating principles. If positive pore pressures are of primary interest, the piezometers should have a low air entry filter and be intended solely for positive pressures. The piezometers with high air entry filters should be redundant piezometers, intended to solely measure negative pressures.

25.9 Measurement of Tilt and Inclination in Slopes, Walls, and Structures

Tilt and inclination measurements are important in many geotechnical applications. They are used to assess horizontal or vertical movements of structures, and soil and rock masses by simple trigonometry from angle change measurements. A tilt sensor measures a rotation angle with reference to the direction of gravity. The simplest tilt sensor that has been historically used is a pendulum. Nowadays, tilt sensors are based on electronic components, but they still measure angles with reference to gravity.

Tilt, in the sense of angle change, is measured by tiltmeters which are usually installed on wall surfaces or bridge piers in situations where rigid block movement with a single rotation point is assumed. If that assumption is valid, then a single tilt measurement should be sufficient to characterize the whole wall or pier horizontal displacement, or vertical displacement if mounted on a horizontal surface such as a bridge girder or a railroad track.

In situations where a series of tilt measurements are required to obtain a full horizontal displacement profile, such as in soil or rock slopes, the measurements are referred to as inclinometer measurements and their objective is usually to establish a horizontal displacement profile from the bottom of a borehole to the surface by a series of tilt measurements at regular (e.g., 0.5 m) intervals conducted with an inclinometer probe or similar device.

Originally, the tilt sensors used in tiltmeters and inclinometers were based on sensing principles such as potentiometers, servo-accelerometers and electrolytic sensors. These sensing principles are still used occasionally, but the majority of modern tiltmeters and inclinometers are based on Micro-Electromechanical Systems (MEMS) accelerometers. These sensors are micromachined on silicon wafers by microfabrication techniques similar to the ones used for semiconductors. They have excellent metrological properties and being solid-state, they are robust against shocks and have reduced temperature sensitivity. They typically have a repeatability of ± 0.002 degrees (e.g., a precision of 0.03 mm/m). This represents a precision of 0.015 mm on the base length of 0.5 m of an inclinometer probe or ShapeArray sensor, or a precision of 0.06 mm on the 2 m base length of an In-Place Inclinometer (IPI). Accelerometers for tilt and inclination measurement were until recently available only in uniaxial versions, so that two separate sensors were required in biaxial inclinometer probes and tiltmeters. Most manufacturers are moving towards the use of triaxial accelerometers that allow the versatility of making biaxial measurements in two horizontal directions or settlement measurements when the survey is conducted horizontally.

Most modern tiltmeters and inclinometers have a digital output since microcontrollers can be easily (and affordably) incorporated in each sensor to conduct the analog to digital conversion and store calibration

factors. This is made even easier by the MEMS accelerometer sensors that are small size and lend themselves to be mounted on the same electronic board as the microcontroller. This digital output allows strings of bussed digital sensors to be made such as the In-Place Inclonometers and ShapeArrays discussed below. In these cases, sensors that have a digital address are mounted along a single signal cable, typically using a RS-485 interface.

25.9.1 Tiltmeters

Tiltmeters are tilt sensors installed to a surface with a vertical or horizontal mounting plate. Tiltmeters can be uniaxial or biaxial, e.g., measuring angle change in one direction or two perpendicular directions which are usually referred to as “A” and “B” directions. The choice of a vertical or horizontal mounting plate is just for installation convenience. In both cases, the measured angle change is the same.

Tiltmeters usually incorporate a temperature sensor. Plotting angle change and temperature on the same graph can be very useful for interpretation, as diurnal thermal changes can cause substantial dilation of the concrete, steel, or rock surface on which the tiltmeter is mounted. Thermal effects can be accounted for by plotting angle change vs temperature of the tiltmeter. This is done in the initial stages of the monitoring program while the structure has not yet been subject to external changes that would make it move or otherwise react and obtaining the slope of the curve that can then be used as a temperature correction factor (baseline) of the angle change readings for subsequent measurements. However, thermal effects on the structure are often not immediate and can lag a few minutes or hours because of the time required for the temperature change to reach the interior of the structure. To account for this, Schuyler and Gularte (2000) suggest a formula to model diurnal structural motions with a sine wave function that includes parameters to adjust the amplitude, phase, and symmetry of the waveform and they have applied it successfully to a bridge deck vertical deflection instrumentation program where a maximum of 2.54 mm was allowed during each episode of compaction grouting of the bridge pier foundations.

25.9.2 Inclonometers

Inclinometer measurements are used to establish horizontal displacement profiles using traversing probe sensors or in-place sensors in access tubes called inclinometer casings, typically installed vertically, and grouted in boreholes below the ground surface (Machan and Bennett, 2008). Alternatively, although less commonly, the inclinometer casing can be installed at an angle or horizontally, and the resulting measurements will provide a vertical settlement profile along the length of the installation. Horizontal installations are discussed in Section 25.10.5.

A typical inclinometer system is shown in Figure 25-2. Its main components are an inclinometer probe sensor, a signal cable, and a reel. The probe sensor implies a manual reading method by which the inclinometer probe is lowered to the bottom of the inclinometer casing by an operator and is raised by increments of typically 0.5 m (or 2 ft. for imperial probes) to read an inclination angle at each elevation. The probe incorporates wheels at a spacing of 0.5 m (or 2 ft. for imperial probes), so that the entirety of the inclinometer casing will be covered by successive tilt measurements and the probe is connected to a signal cable. The signal cable has internal reinforcement so that it will not stretch with time, and markings every 0.5 m (or 2 ft. for imperial probes) so that the operator will know the depth. At each increment, the probe is allowed time to stabilize before taking a reading and proceeding to the next increment. The surface end of the cable is reeled back on a cable reel which may incorporate a Bluetooth radio that transmits measurements to a Field PC or a mobile device that will store data and process it. This allows the operator to compare the displacement profile of the borehole to the baseline survey stored in the device and identify potential measurement errors.



Figure 25-2 Digital MEMS Inclinometer System

Inclinometer casings (Figure 25-3) are generally made of ABS which has better mechanical properties than PVC. The casing incorporates two sets of longitudinal grooves 90 degrees apart which are used as tracks for lowering and raising the inclinometer probe while conducting the measurement survey. In most situations, the second set of grooves will not be used in the survey. It was originally intended to conduct surveys in the perpendicular direction (referred to as 'B' direction) from the main direction of interest, but modern probes are now biaxial so that the second survey at 90 degrees is not required. However, a second survey is always conducted in the main set of grooves by rotating the probe 180 degrees to average the readings and eliminate possible systematic bias in the probe or slight difference at the measurement location. Such bias may be due to calibration shift with time, wheel degradation, hasty readings, and other causes. The magnitude of the bias can be visualized by the operator by verifying the checksum errors at every elevation. These checksums should be constant along the whole profile. Spikes in checksums probably mean that one of the two readings at the given elevation has been erroneous for some reason and the measurement at that depth should be repeated.



Figure 25-3 Inclinator casing

Inclinometer casing is typically available in two standard sizes: 70 mm OD/59 mm ID and 85 mm OD/73 mm ID. In general, the larger size casing should be used when large horizontal displacements are expected as this will extend the useful life of the casing installation until excessive deformation impedes probe ingress past the deformation zone.

Most installations in soil will involve drilling using either HWT-cased mud rotary, hollow or solid stem augers or sonic drilling. Drilling in rock will mostly be in PQ diameter when using diamond drilling, or similar diameters when using tricone or down-the-hole hammer drilling. Both HWT and PQ will provide a 123 mm diameter borehole which is adequate for both inclinometer casing diameter. In some cases, HQ (96 mm borehole diameter) can be considered for the smaller diameter casing.

The borehole should be drilled as vertical as possible. A verticality of ± 1 degree is ideal, whereas ± 3 degrees can be tolerated. Although inclinometer systems can generally operate at ± 30 degrees, a vertical borehole reduces cross-axis effects in the probe measurements. Tolerances for verticality should be checked and discussed with engineers on a project specific basis.

Once the borehole is drilled, the inclinometer casing is lowered to the bottom of the borehole often with a small diameter (19 to 25 mm) tremie tube attached to its side for grouting. Individual casing sections are generally 3.05 m (10 ft) long and casings are linked by a simple snapping mechanism at the couplings fitted with an O-ring for proper sealing. Since the couplings are typically the weak point of an installation where casing become highly deformed by shear movements, ABS solvent cement should be used for installation at depths greater than 50 m to strengthen the joints. Duct tape should be wrapped around each coupling even for shallow installations to provide extra stability and as an additional seal.

Prior to grouting the casing, a strategy must be established to ensure that it will not tend to become buoyant. In all cases, the casing should be filled with water, but this will not be enough to prevent buoyancy as the specific gravity of a typical cement-bentonite grout is in the range of 1.3 as compared to 1 for water. Alternatives for controlling buoyancy include a casing anchor at the bottom of the installation or lowering a weight inside the casing, such as a chain or drill rods, while taking care to not damage the bottom cap. Adding weight temporarily to the top of the casing is not recommended as it will cause the casing to bend (undulate) along its length and the quality of inclinometer measurements can be compromised.

The grout mix is generally the same as one of the two grout mixes presented in Table 1 for fully grouted piezometer for either soft soils or medium to hard soils. Even for rock installations, the latter grout mix is also suitable. For deep installations, staged grouting may be required to avoid subjecting the casing to excessive radial pressure due to the weight of the grout column. A good assumption for the maximum allowable differential radial pressure between the water inside and the grout outside of the casing is in the range of 1.5 MPa for the smaller diameter casing and 1 MPa for the large diameter casing. Staged grouting can also be an alternative for anchoring the bottom part of the casing to counter buoyancy.

It is a good practice to conduct a baseline survey of the inclinometer once the grout has sufficiently cured, typically after 24 hours. Full curing is certainly longer than 24 hours, however the position of the casing in the borehole should already be stabilized after 24 hours for all practical purposes. Proper grout curing can be confirmed by putting a grout sample in a plastic container and by refrigerating it to simulate cooler ground temperatures and to examine it after 24 hours to confirm that the grout has become sufficiently hard.

Because of the importance of the baseline survey to which all subsequent surveys will be compared, it is also good practice to do a second baseline survey immediately after another 24 hours to confirm the validity of the first survey. If the second survey does not match the first one, a third one should be performed. It

would not be a good practice to average two surveys with the preference being to identify a proper (repeatable) survey as the baseline. If possible, the same inclinometer probe and cable should be used in subsequent surveys to increase system accuracy, especially if those components are older as there may have been instrument deterioration (bias) such as probe wheel degradation, calibration, cable stretching and others. Care and proper maintenance/storage of the inclinometer system is of paramount importance to ensure accuracy and repeatability of the inclinometer surveys. When all procedures are followed and surveys are done with sufficient care, the system accuracy of a survey should be within 2 mm per 25 m of casing length which is the typical manufacturer specification for inclinometer systems.

Figure 25-4 shows the typical inclinometer profile graphs that are usually produced using a dedicated inclinometer software provided by manufacturers:

- The cumulative displacements profile, e.g., the difference between the current profile and the baseline profile, is displayed at various dates over the full length of the inclinometer survey. This allows the progression of the horizontal displacements in two perpendicular directions called A (parallel to the expected direction of movement) and B (perpendicular to the expected direction of movement) to be viewed. The depth at which a shear displacement is occurring (~ -35m) and the magnitude of the displacement (18 mm) are visible on the profile. It is worth noting that small displacements are occasionally observed on inclinometer profiles in the upper few metres near the surface, but they are usually of little consequence (to the depth of movement being monitored) since these are often a result of surface disturbance and/or environmental effects from frost for example.
- The incremental displacement profile that displays the difference between the current profile and the baseline profile at each depth increment over which the change of inclination has been measured, typically 0.5 meter, is helpful to identify the movement depth.
- The absolute profile that displays the true shape of the inclinometer casing which is helpful to confirm that the recommended verticality criteria of +/- 1 degree has been met and that the casing doesn't have excessive zones of undulation which would indicate that the casing has been held from the collar to compensate upward buoyancy forces, instead of being held from its bottom.
- The checksum profile which is helpful to confirm that they are sufficiently small and constant over the full length of the profile, preferable within a range of +/- 2 mm at each depth along the profile with no spike value outside that range. A single value outside the range would mean that one of the two measurements made at a given depth is wrong, and depending on the magnitude of that spike value, this could be enough to create an apparent displacement that does not exist at all in the field. In that case, the measurement at that depth should be repeated.

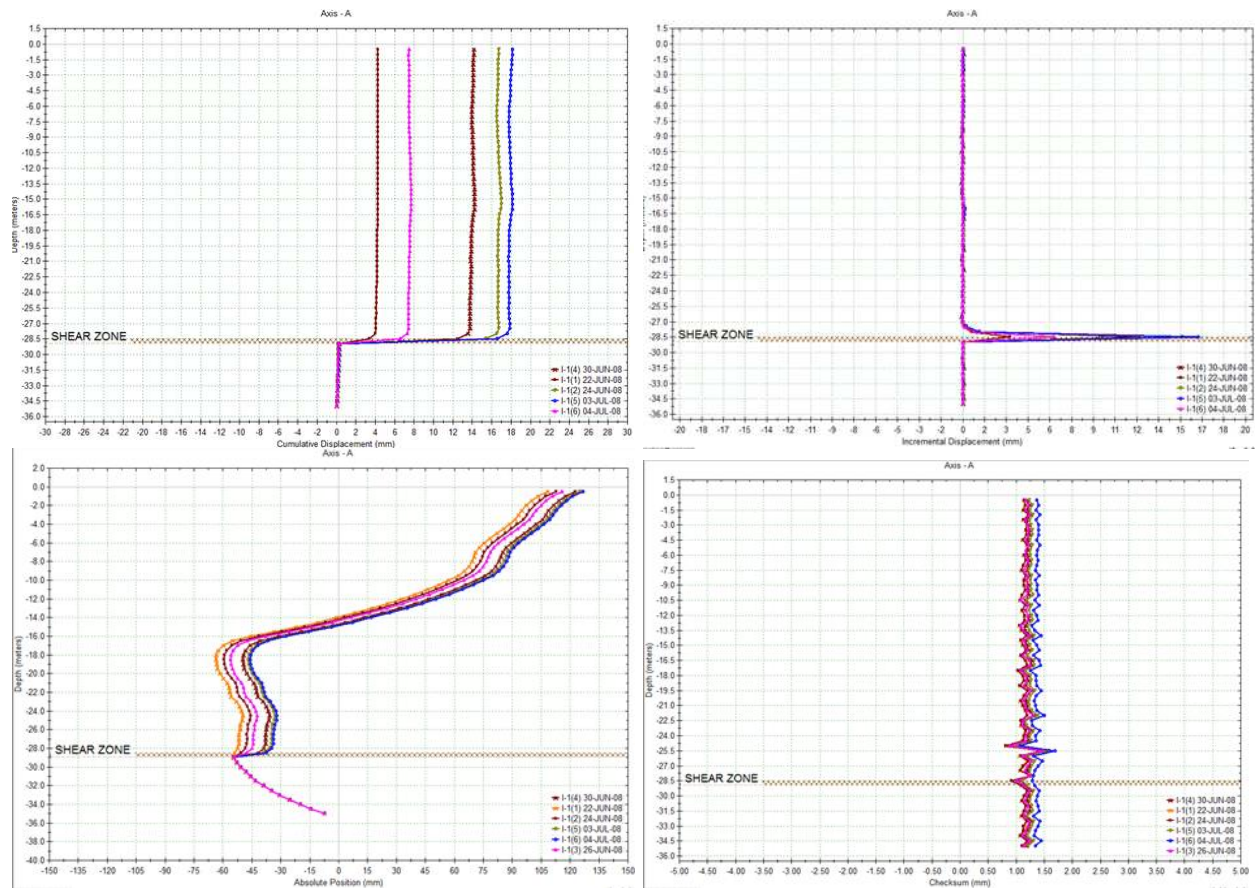


Figure 25-4 Typical inclinometer profiles (Cumulative displacement (top, left), Incremental displacement (top, right), Absolute profile (bottom, left), Checksums (bottom, right))

25.9.3 In-Place Inclinometers and ShapeArrays

In situations where there is a desire to automate inclinometer measurements, In-Place Inclinometers (IPIs) can be used. They consist of a series of inclinometer sensors linked together and lowered in the inclinometer casing and left in place long term (Figure 25-5). The length of the sensors (e.g., the spacing between wheels) can be modified by using extension rods with typical lengths of 0.5, 1, 2 or 3 m. The shorter spacings such as 0.5 or 1 m may be required if the anticipated displacements will be large and especially, if displacements are anticipated within a specific zone. The longer spacings of 2 and 3 m are suitable when anticipated displacements are relatively low or when displacements are expected to occur gradually over a considerable length of the inclinometer casing. The objective of selecting an appropriate length is to be sure that the sensor extension rod will not contact the internal casing wall which would affect the readings. Excessive casing deformation combined with excessively long sensors could even impact the ability to recover the sensors as is sometimes done at the end of monitoring projects.

Figure 25-6 illustrates the various configurations for installation of IPIs. From left to right, they can be installed vertically along the full length of the casing to monitor automatically the full horizontal displacement profile of the borehole, in which case the selected sensor length will generally be between 1 to 3 m, or a string of shorter sensors, usually 0.5 m can be lowered to a specific depth where the presence of the major zone of movement has already been identified by previous manual surveys, or they can be installed horizontally to measure vertical settlement.

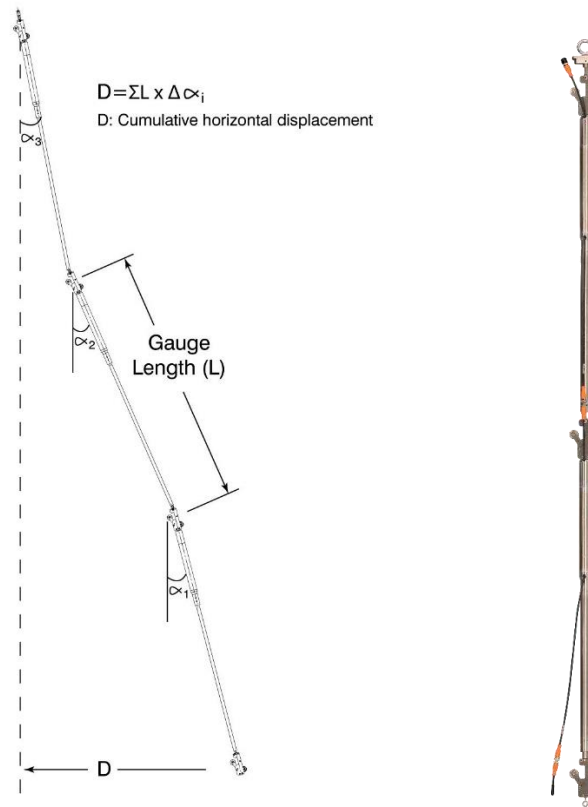


Figure 25-5 In-Place Inclinator sensors

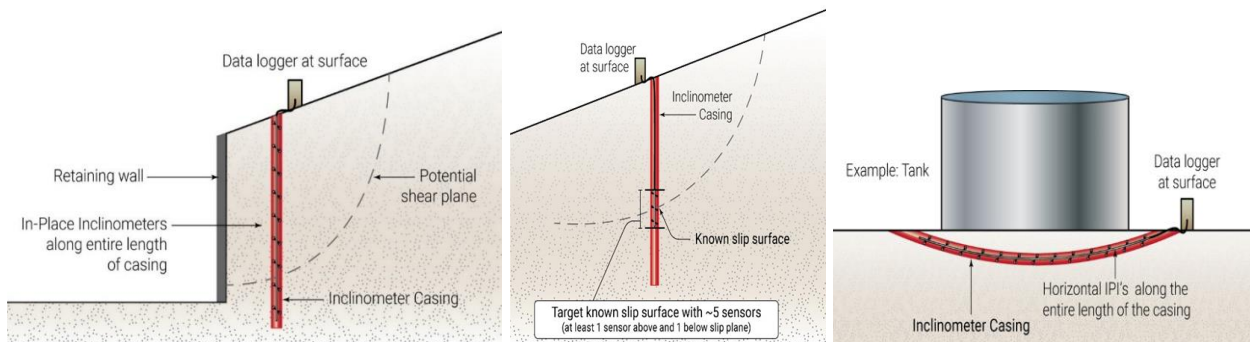


Figure 25-6 Various configurations for installation of In-place inclinometers

A variation of IPIs is the ShapeArray which consists of a number of 18 mm diameter sensorized steel segments, generally 0.5 m long, each incorporating a triaxial accelerometer. These are pre-assembled by the manufacturer and shipped to site on reels. The assemblies are uncoiled from the reel directly into the borehole. ShapeArrays can be installed vertically and horizontally or as a vertical arc to measure convergence in a tunnel or for similar applications. The ShapeArray can be installed in a small diameter conduit (27 mm), a larger diameter PVC pipe, or in existing inclinometer casing of both 59 mm ID and 73 mm ID. When installed in large diameter PVC pipes or inclinometer casings, the ShapeArray segments are pre-compressed such that they zig zag and the flexible joints between the segments contact the casing wall every 500 mm. A special casing collar keeps the ShapeArray pre-compressed. Silent segments can be incorporated in the string of active sensors to concentrate monitoring in zones of interest, and fibreglass extensions installed near the collar can accommodate last minute variations in the borehole depth. The zig

zag fashion allows the ShapeArray software to calculate the centroid of the pipe (at the time of install) and any deviations from that centroid as movement occurs.

In principle, both In-Place Inclinerometers and ShapeArrays measure the same horizontal displacement profiles in the ground. However, certain situations can make one or the other preferable as listed below. Figure 25-7 and 25-8 show comparative installations of a ShapeArray and an In-Place Inclinerometer. Figure 25-7 (left photo) shows the zig zag pattern of the Shape Array in the inclinometer casing.



Figure 25-7 Installation of a ShapeArray in a vertical inclinometer casing



Figure 25-8 Installation of an In-Place Inclinerometer in an inclinometer casing

25.9.3.1 Considerations for Selecting In-Place Inclinerometers

- Maximum total expected displacement along the full profile: about 300 mm (max. 50-75 mm for distinct shear zones),
- Mainly used when the objective is to confirm that there is zero or limited displacements,
- Can be more economical when sensor lengths are 2 or 3 m,
- Allows to easily target a specific zone in the borehole by suspending a string of sensors on a steel cable, and
- More modular allowing mix and match sensor lengths - can be adjusted on site.

25.9.3.2 Considerations for Selecting ShapeArrays

- Maximum total expected displacement along the full profile: up to 1000 mm and more,
- Suitable for both small and large displacements,
- Suitable for both situations: confirm that there is no displacement or monitor very large displacements,
- Similar or better system accuracy as In-Place Inclinerometers,
- Easier and quicker to install, especially for longer Shape Arrays,
- No connecting of sensors in the field. Pre-assembled array of custom length in factory, and
- Can be used to extend life of inclinometer casings that can no longer be read with conventional probe inclinometers.

25.10 Measurement of Settlement, Heave and Differential Settlement

In addition to using geodetic measurements discussed in Section 25.17, ground settlement and heave are commonly measured by one of the five methods described below. As a rough differentiator, geodetic measurements are often preferred for measuring displacements of buildings and other steel and concrete structures, while methods using geotechnical instrumentation are preferred for measurements in soils and rocks. There are however situations where both geodetic and geotechnical instrumentation methods complement each other on construction projects.

25.10.1 Probe Extensometers

This method involves an access tubing installed in a borehole with reference targets at intervals. The depth to these targets is measured by a probe lowered in the borehole. The access tubing can be a small diameter PVC pipe (typically 25 mm diameter) or it can be an inclinometer casing that will have a dual use for measuring horizontal displacements with an inclinometer probe and vertical displacements with the probe extensometer. The reference targets are generally magnet rings with a reed switch probe mounted at the end of a graduated tape that closes a contact when it is raised into the tubing and intersects the magnetic field (Figure 25-9).

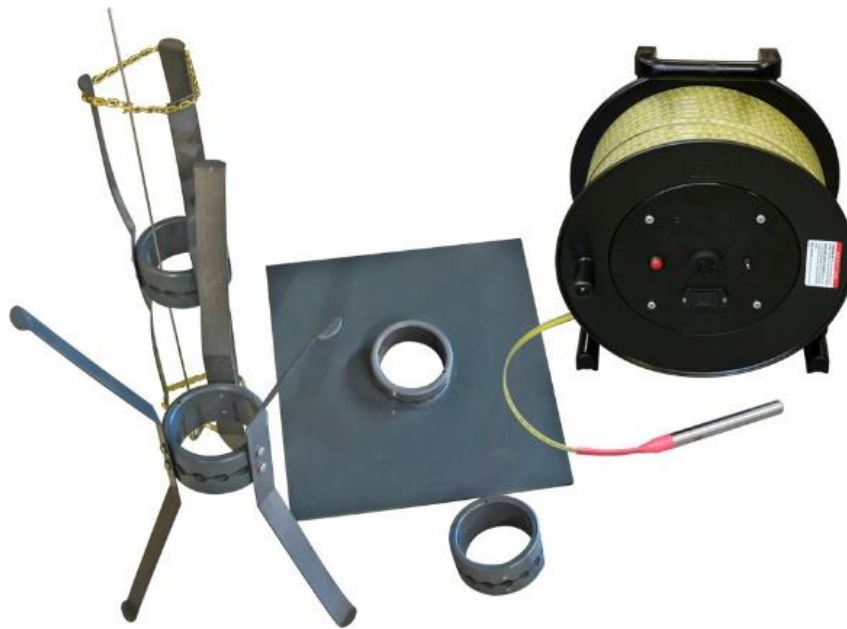


Figure 25-9 Probe extensometer, magnetic settlement plate and spider anchors

If large compression or heave is expected, the access tubing requires telescopic couplings at intervals. The magnet rings can be mounted on those couplings or can be separate spider targets or plate targets. Spider targets are used when the tubing is installed in a borehole whereas plate targets are used in embankments as they are raised.

25.10.2 Borehole Extensometers

A multi-point borehole extensometer, often called a MPBX (Figure 25-10) consists of PVC-sheathed rigid steel rods (typically 6.35 mm diameter) or flexible fiberglass rods (typically 4.75 mm diameter) terminated by an anchor at the bottom of a borehole. The anchors can be short rebars for anchoring by grouting, or hydraulic Borros anchors which are steel prongs that are deployed by hydraulic pressure, or other specialized anchors.

Settlement or heave displacements are measured at the surface on a collar anchor which acts as a reference plate if displacements are measured by a depth gage. Alternatively, an electrical head with vibrating wire or potentiometric displacement transducers can be mounted on the reference plate. The design of the electrical head generally allows re-adjusting the range of the displacement transducers when they are close to their maximum.

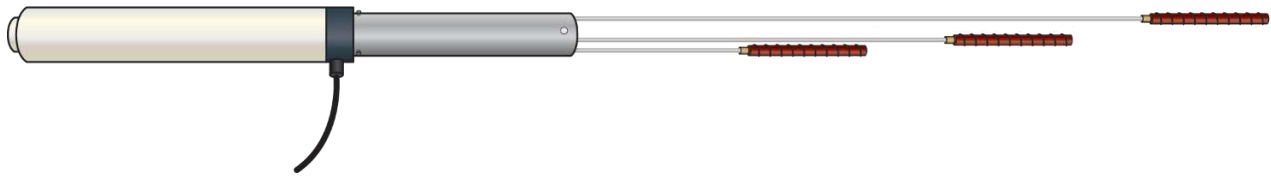


Figure 25-10 Multi-point borehole extensometer (MPBX) with electrical head

Where substantial settlement is expected, a telescopic PVC sheath should be used over the steel or fiberglass rods and the grout should be compatible with the stiffness of the ground. Grout mixes similar to those described in Table 25-1 for fully-grouted piezometers should be used.

25.10.3 Liquid Settlement Cells

Liquid settlement cells work on a principle of hydraulic pressure acting between a fluid reservoir mounted on a fixed point away from the settlement zone and a settlement cell consisting of a low-range pressure electric transducer (vibrating wire or piezoresistive) mounted on a small reference plate and connected by two hydraulic tubes to the reservoir. When displacements occur, the hydraulic head acting on the pressure transducer changes in direct proportion to the settlement or heave movement as illustrated on Figure 25-11. The hydraulic fluid is typically a 50:50 mix of de-aired water and glycol. The lowest range pressure transducer used in practice is a 70 kPa corresponding to a fluid head of approximately 7 m. Based on a typical accuracy of $\pm 0.1\%$ F.S. and precision of $\pm 0.025\%$ F.S. for a vibrating wire pressure transducer, the system will have an accuracy of about ± 7 mm and a precision of about 1.75 mm. Considering these accuracy and precision numbers, liquid settlement cells are typically only used when substantial settlements are expected.

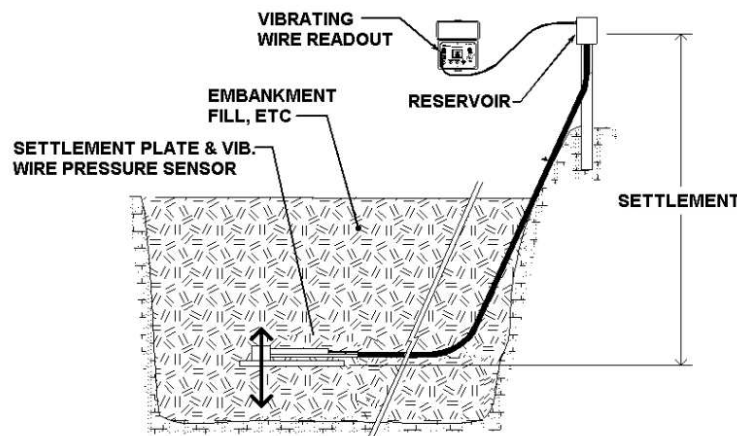


Figure 25-11 Schematic of a Vibrating wire liquid settlement cell system

Since the distance between the settlement cells and the reservoir can be long, sometimes exceeding 100 meters and even up to a few hundred metres, the hydraulic tubes should be installed in a trench in such a way that they will not be compressed or kinked. Good practice is to use a protective PVC or corrugated polyethylene protection pipe around the tubes and the tubes should be installed horizontally or with a slightly upward slope towards the reservoir to avoid trapping air bubbles which would affect accuracy of the readings.

Experience shows that air bubbles develop gradually in the tubes and that they must be purged periodically, possibly every year or few years or when readings start to become unstable. Purging the water-glycol fluid must be done with new de-aired water glycol and a vacuum pump. This can be a cumbersome operation, although portable de-airing and purging units are available to more easily do this operation in the field.

Depending on the design of the reservoir, barometric pressure changes will affect the readings of the pressure transducer. For this reason, readings should be corrected for barometric pressure changes using a barometer. An alternative correction method is to use a reference settlement cell installed on stable ground and connected to the same reservoir as the active cells. Correction will then be applied by subtracting the readings of the reference cell. This method of correction presents an additional advantage as it accounts for temperature related density change of the hydraulic fluid which also affects readings.

25.10.4 Differential Liquid Settlement Cells

Differential liquid settlement cells, as illustrated in Figure 25-12, work on a similar principle of hydraulic pressure as liquid settlement cells except that the design incorporates a differential pressure sensor. The hydraulic fluid pressure acts on both sides of the pressure transducer, thereby allowing a lower range pressure transducer to be used, typically less than 10 kPa, so that the total measuring range can be 1 m and the system will then have an accuracy of about ± 1 mm and a precision of about 0.25 mm (assuming the same full scale accuracy and precision noted in Section 25.10.3) for the pressure transducer. Considering this accuracy and precision, differential liquid settlement cells are typically used if small differential settlements are expected, for example when installed along a wall of a building or in an underground parking or along a tunnel wall or a bridge girder.

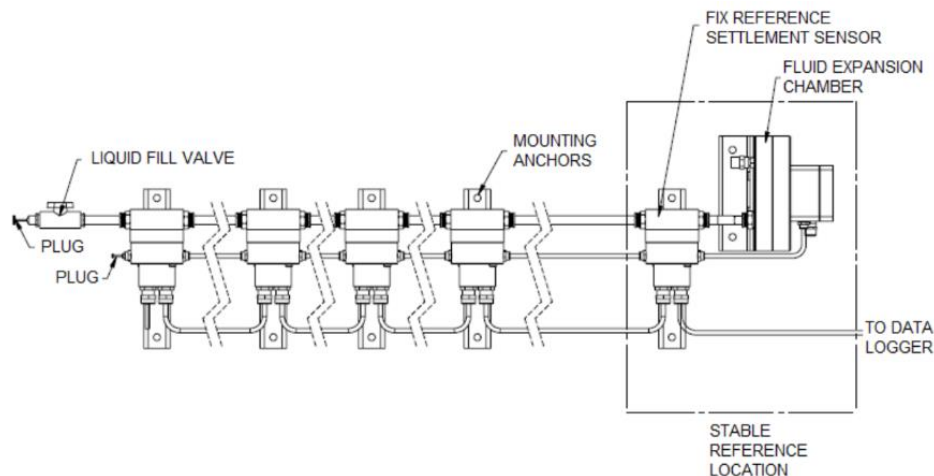


Figure 25-12 Schematic of a Differential liquid settlement cell system

25.10.5. Horizontal inclinometers

Inclinometer casing can be installed horizontally to measure a settlement profile under earth fills (e.g., highway embankment), at the base of excavations, or beneath structures or tanks. A common application is measuring foundation settlement for large Liquefied Natural Gas (LNG) tanks when they undergo a hydrostatic proof test prior to putting them in operation.

The casing can be installed in a trench before construction or in a horizontal borehole drilled by directional drilling. The casing can also be installed with a radius of curvature to allow installation under an existing

highway (for example). Surveys are conducted by pulling the probe in the casing and taking a measurement every 50 cm. Once the first pass is done, the probe is pulled back in the casing using a steel cable attached to its other end, reversed and passed through the casing again. The probe is reversed end to end, not rotated axially 180° as in a vertical inclinometer survey. The readings taken represent the vertical displacement defined by the product of the reading interval (0.5 m) and $\sin(a)$ where “a” is the angle change between the horizon and the longitudinal axis of the probe.

25.11 Measurement of the opening of Cracks

Crack openings are measured with crackmeters which are displacement transducers mounted on anchors installed on concrete, rock, or soil surfaces or inside those materials. When installed in concrete, rock, or soil, they are usually installed in boreholes, and the crackmeter is then called a borehole extensometer as described in Section 25.10.2.

25.11.1 Crackmeters and Jointmeters

Crackmeters typically have a measuring range of 25 mm to 200 mm which is sufficient for most crack opening measurement applications. The anchors are usually 6 or 9 mm diameter short rebars, 10 to 15 cm in length, that are grouted in small diameter holes. Anchors should be grouted at least a few centimeters from either side of the crack to be measured. The displacement transducers are usually of the vibrating wire or potentiometric type (Figure 25-13).



Figure 25-13 Vibrating wire crackmeter

Crackmeters can also be used to measure the opening of construction joints in concrete, for example between concrete deck sections of bridges or construction joints in concrete slabs or blocks. If installed at surface, they will usually still be called crackmeters, while they are called jointmeters in the case of special models to be embedded in concrete.

25.11.2 Soil Extensometers

Soil extensometers (Figure 25-14) are embedded in soil. Their design is similar to that of a crackmeter but with a longer distance between ends, typically 1 to 3 m, although 6 m can also be considered. They can also be installed in series to span a longer measuring length, for example a zone of tension cracks on a slope. Individual soil extensometers have a measuring range of 50 to 300 mm, either of the vibrating wire or

potentiometric type. They have two large diameter anchor discs at both ends, so that the extensometer can measure both extension and compression in the soil. At the time of installation, the extensometer can be set at the middle of its measurement range or be set so that it will measure mostly extension or compression, depending on the anticipated nature of the movement.

Soil extensometers are usually installed horizontally and at shallow depths in soil trenches unless they can be installed during construction of embankments as it is raised. If a vertical installation is needed, a borehole extensometer would likely be selected.



Figure 25-14 Vibrating wire soil extensometer

25.12 Measurement of Strain and Load in Concrete, Steel and Anchors

Strain in concrete, steel beams, steel rebars and rod or stranded anchors are measured with strain gauges. Loads on soil and rock anchors and on piles during load tests are measured with load cells.

25.12.1 Strain Gauges

A strain gauge is a surface-mountable or embeddable sensor that measures strain, which is expressed in micro-deformations, e.g., the ratio $\Delta L/L$, with L being the base length of the strain gauge. Strain is therefore nondimensional being a ratio which expresses a change of length over a certain length. The word “micro” in micro-deformation means strain is expressed in a number multiplied by 10^{-6} . For example, 1 micro-strain means 0.000001. Strain can be converted to stress using Hooke’s law assuming the load is within the elastic zone. Beyond the specified yield strength (or grade), the stress/strain relationship for the material being monitored should be used.

Strain gauges can be of the resistive type bonded to steel or concrete and read in an electrical circuit which is called a Wheatstone bridge. Resistive strain gauges are used extensively in short term and some long-term indoor applications with short lead wires. However bondable resistive strain gauges require considerable skills for installation and waterproofing and may not be suitable for outdoor applications.

Vibrating wire strain gauges are generally preferred to resistive type when outdoor use and long-term reliability are required. They consist of a vibrating wire mounted between end anchors. The end anchors are arc-weldable or spot-weldable where mounted on steel or groutable if mounted on concrete.

The measuring range of vibrating wire strain gauges is typically 3000 micro-strains and the repeatability is in the order of 1 micro-strain or better. Some models (especially the embeddable type) are factory-set at the mid-range while some models are set at the time of installation. Depending on whether the anticipated strain is mostly compressive or tensile, the strain gauge will be set accordingly by the user.

When strain is to be monitored in concrete, an embeddable vibrating wire strain gauge as illustrated in Figure 25-15 can be used. It is similar to the surface-mounted model, but the end anchors are replaced by

two circular end flanges. This type of strain gauge can be directly embedded in concrete, for instance by tying loosely with steel wires the two ends of the strain gauge to steel rebars to ensure that they are oriented in the proper measurement direction.



Figure 25-15 Vibrating wire embedment strain gauge

25.12.2 Load Cells

A load cell is a steel cylinder, annular or solid, instrumented with internal strain gages, generally of the vibrating wire or resistive type. Annular load cells as illustrated in Figure 25-16 are used to measure load on soil and rock anchors and solid load cells are used to measure load on piles during static pile load tests. Thick top and bottom load platens are essential for hollow load cells to ensure uniform load distribution in the body of the load cell and minimize errors due to eccentricity. Sellers (1994) recommends that platens should be ground smooth, flat, and parallel and with the minimum thicknesses listed in the table below to minimize bending and warping of the platens under the considerable loads concentrated on the cylindrical perimeter of the load cell.

Load Cell Capacity (KN)	Minimum Top and Bottom Platen Thickness (mm)
750	40
2000	65
3500	75

Table 25-2 Relationship between load cell capacity and minimum thickness of top and bottom load platens

Sellers (1994) discusses the frequently observed mismatch between load measured by the calibrated hydraulic jack used to pretension ground anchors and annular load cells installed between the jack and the anchor head plate. He concludes that if the jack's piston diameter is smaller than the diameter of the load cell, the load cell will tend to over-register, while a jack with a bigger diameter than the load cell will cause the load cell to under-register. The effect is greater if the platen between the jack and the load cell is thinner, thereby validating the minimum platen thicknesses listed above. He points out an additional consideration that friction losses within the hydraulic jack can cause under-registering of jack load indications by as much as 15%.



Figure 25-16 Annular Vibrating wire load cell with top and bottom load platens

25.13 Measurement of Earth Pressure

25.13.1 Total Pressure Cells

A total pressure cell (Figure 25-17) measures earth pressure in soils or contact pressure between a soil structure and a structure (e.g., a bridge abutment). It is comprised of a hydraulic pressure pad, generally 20 to 30 cm in diameter, and a pressure transducer, either vibrating wire or piezoresistive, connected to the pressure pad by a small diameter tube.



Figure 25-17 Total pressure cell with vibrating wire pressure transducer

The pressure pad is made of two circular steel plates welded at their periphery. The thin gap between the steel plates is filled with a de-aired hydraulic fluid, so that earth pressure changes acting perpendicularly to the pressure pad will be hydraulically transmitted to the pressure sensor. To accurately measure pressure acting in the soil, the pressure cell, and the backfilling around it, must have a stiffness similar to the enclosing soil. It should not be a rigid inclusion in the soil, in which case it would overestimate pressure, and it should not be a soft inclusion and therefore underestimate pressure. For this reason, installation of the pressure cell must be made by preparing a suitable base under the cell, and backfilling above it and compacting soil to a similar density as the surrounding soils.

25.13.2 Push-In Pressure Cells

A push-in pressure cell (Figure 25-18), also called Spade cell, can be pushed in soils using a drill rig. Tedd, P. et al. (1990), Richards, D.J., Clark, J., Powrie, W. and Heymann, G. (2007) and Richards, D. J., Powrie, W., Roscoe, H. and Clark, J. (2007) discuss applications, installation, and data for these cells. Push-in pressure cells measure horizontal in situ stress in soils and changes in total pressure due to soil improvement procedures such as stone columns, vacuum consolidation, or preloads. The pressure cell also incorporates a pore pressure transducer to determine effective stresses.



Figure 25-18 Push-in pressure cell

25.14 Measurement of Temperature

Temperature cables are used to monitor temperature along horizontal or vertical profiles. The sensor type is usually an NTC (Negative Temperature Coefficient) thermistor or another temperature sensor type such as a RTD (Resistance Temperature Detector) or a thermocouple, connected and sealed at the end of a single cable or at various positions along the length of a multi-conductor cable. In that latter case, the temperature cable will be called a thermistor string. Thermistor strings are limited to a fixed number of measurement points along the multi-conductor cable, typically to a maximum of 24 points. In recent years, digital temperature cables have also been introduced where each sealed node has a thermistor and a micro-controller so that a much larger number of bussed temperature measurement points can be assembled along a 3 or 4-conductor cable. Digital temperature cables can have more than 100 measurement points and can be up to 1 km long.

Thermocouples can also be used for temperature measurement, especially when short term measurements are required, such as the curing temperature of concrete. For longer term applications, thermistors or digital cables are generally preferred as they will be easier to read either with portable readouts or dataloggers.

Measurement of temperature is often required for construction projects in zones of permafrost (Hoeve and Zhang, 2019), in zones that experience seasonal freeze/thaw or where artificial freezing is used for stabilizing ground or keeping groundwater out. As an example, when refrigerated building foundations are used, it becomes important to monitor temperature long-term and at depth and confirm the permafrost does not thaw due to heat transfer from the building or from exposure to surface temperatures during the warm season, which could lead to differential settlement. Highway embankments built over permafrost can be subject to slumping due to permafrost thaw, leaving the road surface cracked or deformed and in need of regular repair (McGroarty and Billing, 2019). Temperature monitoring is commonly used for monitoring performance of mitigating features, such as thermosyphons, used in design of new or rebuilt sections of highways.

Artificial ground freezing is becoming more common as a means of ground shoring for excavations, shafts, and tunnels (Khawaja et al., 2019). It can also be used for containing contaminated groundwater or in reverse, keeping groundwater out of an excavated area. Temperature sensors are used to monitor the growth of the freeze bulbs within the ground prior to the start of an excavation. Temperature monitoring can also be used for several other applications such as monitoring seepage in embankment dams and the performance/operation of geothermal systems.

25.15 Automated Data Acquisition Systems and Telemetry

Most instruments can be read by portable manual readouts dedicated to the measuring principle of the instrument (e.g., vibrating wire, MEMS, potentiometric). However, manual readings are very time consuming and can be costly when considering the personnel time to conduct the readings. However, one advantage of manual reading, is it allows a visual inspection of the site. Nevertheless, and with the increased availability of automated methods to read instruments, automatic datalogging has become a more common method of reading instruments.

In the context of geotechnical instrumentation, Automatic Data Acquisition Systems (ADAS) can be described as field deployable electronic devices that collect and store measurements from a variety of sensors connected to them by signal cables or by radio (Choquet and Taylor, 2014). One of the advantages of an ADAS is that readings are taken, stored and as an option, transferred automatically to a remote computer without any human intervention, e.g. avoiding the traditional manual readings with portable readouts by a person in the field. Features required for ADAS used in geotechnical instrumentation include:

- Signal management for a variety of geotechnical sensors – e.g. vibrating wire, MEMS sensors with digital output, 4-20mA transmitters, analog output (typically 0-5 or 0-10 Volts, digital output sensors (e.g., RS-485, SDI-12),
- Typically battery-powered with industrial type alkaline or lithium batteries, or lead-acid rechargeable batteries with solar panel,
- Data storage memory in the field device,
- Typically low-throughput communications – e.g. radio, cell modem or satellite, in order to transmit measurements on a regular schedule to a local or remote supervisory office,
- May have local alarm/control capability, and
- A continuum from 1 channel standalone dataloggers to 1000+ channel distributed datalogger network.

With respect to the differences between a datalogger and an ADAS, a datalogger is a single device that incorporates all or most of the above features. An ADAS refers to an elaborate datalogger with at least all the above features and possibly more, such as the ability to be deployed as a network of several units, manage simple or elaborate alarm conditions, and possibly control external devices such as a flashing light or a siren in case of an alarm condition. ADAS may also be capable of sending an SMS or actuating a valve or some similar response to an alarm or threshold condition.

With advancements in technology, the current market offering for monitoring programs for geotechnical instrumentation is now showing small dataloggers incorporating features that were originally only available in ADAS networks. This is evolving the market as the technology becomes more accessible and demands for monitoring increases.

25.15.1 Small Dataloggers

Small dataloggers are typically low-cost standalone battery-powered units packaged in a small enclosure and can read either a single sensor (single channel loggers) or a few sensors, typically 5 or 10 (Figures 25-19 and 25-20). Because of the low-power processors used and the low sampling rates at which they are generally used, they can be powered by small “D” size or similar 3.6V batteries, and yet have an autonomy of at least one year. The operating temperature range of small loggers can be -40° to +80° C., pending frequency of readings and transmission of data and long exposure to very cold temperature. Recent developments allow the integration of a low-power license-free radio chip so that small dataloggers can be integrated into an ADAS network and still be powered by the same small size batteries or solar panels.

These radios can easily transmit over a few kilometres in open areas, and about one kilometer in urban environments.



Figure 25-19 Single-channel datalogger



Figure 25-20 Small count (5-10 channel) datalogger

25.15.2 Integrated ADAS Networks

Large geotechnical projects may require automated monitoring of up to hundreds of instruments. In such cases, it becomes uneconomical and most of the time technically impossible, to route all instrument cables to a single location. Therefore, ADAS design for a large project typically requires optimizing the number

and the location of data acquisition systems, and the connection method between the units to a central monitoring station.

There are basically two options for this scenario. The first is to have a few data acquisition systems at strategic locations and to route instrument cables to those systems which will then incorporate multiplexers to connect multiple sensors. Individual systems will then be inter-connected by radio or each of them will have a cellular modem. However, this scenario has a few drawbacks:

- the required quantities of instrument cable can be very substantial and can become very costly, in addition to the costs involving in trenching and/or conduit to protect the cables.
- Long horizontal surface runs of instrument cables are a risk for damage to instruments and datalogger because high voltage transients due to lightning strikes tend to travel near the ground surface. To avoid these types of situations, a general recommendation is to keep the cable runs to the sensors as short as possible, to increase the number of data acquisition systems in the network and, if possible, to use radio communications to shorten cable lengths.
- Long cables also add more resistance to readings that, in some cases, can impact readings of sensitive instruments.
- Long cable runs often require splicing of cables which, if not performed correctly, can add a source of error if a splice fails.

The second option makes precise use of more radio communications by leveraging the small radio-enabled dataloggers mentioned in Section 25.15.1 and installing those dataloggers close to the instruments, at borehole collars or near concentrations on a few instruments with short cable lengths. The data acquisition network becomes then a hub-to-node network where the hub is a data acquisition system that communicates by radio to the small dataloggers deployed in the field.

There are two types of hub-to-node networks. The first is a network in star configuration by which all small dataloggers communicate with the central data acquisition system and the second is a mesh network by which all small dataloggers can also act as radio relays to transmit their measurements to a network access unit which is often call a gateway. With either option, communication with the central office is from the central data acquisition system or the gateway. Both options would use a cellular or satellite modem, Wi-Fi or direct LAN access if available on the site.

25.16 Software for Data Management, Visualization, Alarming and Reporting

Most dataloggers and ADAS store and transmit measurements in the form of data files that are simple delimited value (e.g., *.csv or *.dat) files. Their content is generally a line for each set of measurements with a time stamp in the first column and the raw measurements of each datalogger channel in subsequent columns. In some cases, additional columns provide measurements converted to engineering units although most users will prefer to work with the raw readings. The csv files can readily be imported into a spreadsheet (e.g., Microsoft Excel), and many users will do the data management and visualization in that software and will build the required visualization templates with graphs of groups of sensors.

Commercially available spreadsheet software may in some cases be insufficient for the requirements of the instrumentation and monitoring project. This may include projects that involve a large quantity of instruments and, or projects that will have a long duration and considerable alarming and reporting requirements. In this case, specialized software programs are commercially available that are dedicated to the management and visualization of instrumentation data. Most of these programs will have the following features:

- **Dashboard:** a dashboard is the first screen accessed by the user and it is a summary screen to present the most important project data, including an overall view of the project and of the location the instruments. Real time values of sensors and sensors currently on alarm can also be displayed,
- **Real Time Display:** it is the current reading of all sensors, usually with a color code (Green: OK, Yellow: Threshold, Red: Alarm),
- **Trend Lines:** plots of sensor reading vs time. They are the most common visualization tool. Scrolling back and forth in time and zooming are important features,
- **Alarm:** new sensor readings are immediately checked for alarm thresholds and alarm messages such as SMS and emails can be sent automatically,
- **Virtual Variable:** used to create new calculated variables that are a mathematical function of one or more sensor readings.
- **Access Controls:** User Profiles with different levels of access for software configuration or simple visualization,
- **File Converter:** ability to include data from most data logging systems, data from spreadsheets and manually recorded data,
- **Correlation (XY) Graph:** to display data pairs to analyze the correlation between two or more sensors. (Ex: piezometric levels vs rain, strain vs temperature),
- **Profile Views:** to view data from profile sensors such as In-Place Inclinometers and ShapeArrays,
- **Reports:** automatic creation of reports of select sensors on specific schedules.
- **Visualization on smartphones and tablets,**
- **Cloud Hosting:** to facilitate access from different users without firewall and other IT restrictions,
- **Language:** support different languages,
- **Import Geodetic data** (motorized total stations) and InSAR data, and
- **GIS and Google Earth:** allow geo-localization of sensors.

25.17 Geodetic Measurements

Geodetic monitoring methods consist of ground-surveying instruments and procedures used to detect horizontal and vertical movements over a series of fixed points forming a survey control network (De Rubertis, 2018). The fixed points can be specially constructed survey monuments or survey points positioned directly on structures. Several references are available on geodetic measurements, including Chapter 6 of De Rubertis (2018) and Walker and Awange (2020).

25.17.1 Total Stations and Levels

Total stations and levels are the main instruments of surveying for geotechnical monitoring. Contract services from survey firms are available in most cities and regions, or an organization can consider having their own surveyors.

25.17.1.1 Total Stations

A total station is an electronic optical instrument used to obtain angle and distance measurements to a specific point, typically a prism or a reflector on a survey rod or attached more permanently to the point that is being measured (De Rubertis, 2018). Robotics allow the station to be operated by remote control and reflectorless technology is available although accuracy and repeatability are reduced. Automatic target recognition is available so that when the total station is pointed near a target/prism, it will determine the offset and correctly point to the exact center of the target/prism. This allows for automated systems where predetermined initial angles are programmed, then the instrument will aim at the targets and do the final correction automatically. Alternatively, the instrument can also scan and look for the target.

A typical application consists in measuring vertical and horizontal angles distances targets at specific locations on a structure or monuments on the ground. These measurements are collected in sets of angles and distances and can be measured multiple times in direct and reverse modes to reduce calibration and reading errors. To obtain settlement measurements, trigonometric leveling can be done to derive elevations. The accuracy of the elevations will depend on how precisely the total station can measure vertical angles, typically 1 to 5 arc-seconds and distances, typically 1 to 1.5 mm + 1 to 2 ppm.

When using a robotic total station installed on a permanent or semi-permanent location, the instrument can be pre-configured and will complete sets of measurements of multiple targets autonomously and on a schedule. Raw measurements of angles and distances are then converted into usable coordinates including vertical settlement. Further analysis may be required to reduce errors and provide true measurements.

25.17.1.1 Levels

A level is an electronic optical instrument used to measure differences in height between two given points with respect to a reference elevation. In general, elevations measured with levels for the purpose of determining settlement will be more accurate than if obtained using total stations. The accuracy of level measurements depends on the quality of the survey and quality of the equipment used and can be in the range of 3 mm, unless extraordinary methods and precise instruments are used, in which case it could approach 1 mm.

Leveling is typically done with a level set up over a tripod. The surveyor sights a staff or rod with a numbered scale or bar code to obtain a reading. Levels do not have the same type of robotization as total stations, although they have also received considerable technological improvements. Nowadays, most levels are of the automatic type and digital levels are also available.

Automatic levels make use of a compensator that ensures that the line of sight remains horizontal once the level has been roughly leveled. The surveyor does not need to re-level the instrument after rotating to measure another point. Digital levels electronically read a bar-coded scale on the staff. These instruments are the electronic version of the automatic level, and they include a data recording capability. The measurements are made to bar-coded rods and horizontal distances and elevation differences are measured. The digital level may also compute and apply refraction and curvature corrections, which can improve considerably the accuracy and reliability of elevation measurements.

25.17.2 Rotating Lasers

Rotating lasers can be an option for settlement measurement of horizontal or inclined surfaces. It consists of a rotating laser mounted on a tripod and a receiver mounted on a graduated rod. The rod can be positioned on previously marked reference points to do a level measurement, similar to a digital level measurement.

25.17.3 Laser Scanners

Laser scanners use a technology called Light Detection and Ranging (LiDAR) . LiDAR can be used for both aerial (UAV: Unmanned Aerial Vehicle) or terrestrial (ground-based) and even submarine.

The ground-based static version is a laser scanner. It uses a motorized laser to scan a surface, generating a cloud of points, each with its x, y and z coordinates with a resolution that can be millimetric. The maximum range of laser scanners can vary between 300 m and 600 m. A good recommendation is to set up certain number of reference targets and to run the scans along the surface from stations that will maintain an incident angle of 45° or less to the scanned points to minimize bounce back errors of the laser (Laefer & Lennon, 2008).

A promising application of laser scanning is for monitoring deformation of Mechanically Stabilized Earth (MSE) or other retaining walls. Adamson et al. (2019) report on the comparison of measurements with a motorized total station and a laser scanner to monitor settlement of a MSE wall in a highway interchange under construction and post-construction. The conclusion of the authors is that laser scanning is suitable

for that application, although the data presented more variability than that from the total station. Their recommendation is to use a redundant method to confirm the laser scanner, while recognizing that the laser scanner provides much more data in a complete 360° space and allows the user to make dual use of the data by providing a complete 3D model of the structure in which the designer can navigate through without the need of being on site and can use the data for monitoring purposes at the same time.

Other applications for laser scanning include monitoring of rock and landslides. Stark. et al. (2021) provide a detailed description of remote sensing technologies, including LiDAR, as well as the other technologies discussed below: GNSS, InSAR, Digital photogrammetry and UAV.

25.17.4 GNSS Global Positioning

Global Navigation Satellite System (GNSS) global positioning is based on satellite navigation constellations. Originally, there was only one constellation called Global positioning System (GPS) (maintained by the US government, but there are now more constellations from various countries: Galileo (Europe), GLONASS (Russia) and BeiDou (China) that also have global coverage. Geo-referenced positions in any weather conditions at any location on the earth can be obtained. The main component is a GNSS receiver which can be installed permanently or semi-permanently on the ground surface or that can be moved as a rover to obtain position readings on various survey points. The GNSS receiver can determine its position if it has a line of sight to an adequate number of satellites (typically at least 3 or 4). Accurate time information is also provided by the receiver.

If used as a standalone unit, a GNSS receiver allows geo-spatial positioning in the order of 1 metre, depending on the constellation. To increase the accuracy of the positioning, one of two reference systems is used. The first is established by placing GNSS receivers on known stable positions and using them as reference points which can produce a 3D positional accuracy in the range of 3 to 5 mm. Generally, the accuracy will be better horizontally than vertically. The second is called Continuously Operating Reference Station (CORS) which consists of reference receivers permanently mounted to the ground on stable monuments connected to the bedrock. These stations are installed and maintained by local agencies and their access is generally available to the public in exchange of a subscription fee. Most Canadian urban areas and areas beyond now have access to CORS services. CORS also provides 3D positional accuracy within a few millimeters.

GNSS accuracy can be improved by using an RTK (Real-Time Kinetic) system, by which a GNSS base station located on a reference point broadcasts its position by radio to a GNSS receiver used in rover mode at various points on an embankment or any structure to be monitored to obtain real time positions at a few second intervals. Positions within about 10 to 15 millimeter can be achieved in this way.

25.17.5 Satellite InSAR

Interferometric Synthetic Aperture Radar (InSAR) is a technology gaining wide acceptance for obtaining information on ground settlement, and to a certain extent horizontal displacement. It is based on the comparison of multiple Synthetic Aperture Radar (SAR) images obtained during each earth orbit by satellite constellations from space agencies. Each image is analyzed to extract the phase difference between an image acquired at a previous date from the same location and the current image and to remove contaminant signals and other interferences and this information is determined for each pixel of the images. The difference in wavelength phase allows the displacement that has taken place between the image acquisition dates to be determined.

Radar can penetrate through clouds and is equally effective at night. Depending on the radar wavelength band used, it can also penetrate vegetation. Current SAR satellites operate within one of the three most

common wavelengths bands: X-Band, C-Band and L-Band. Shorter wavelengths increase the sensitivity to displacements leading to greater precision, while reducing data coverage in vegetated areas. L-Band uses a longer wavelength and often can penetrate heavily vegetated areas. The result is displacements presented visually for each pixel over optical satellite images of the area covered by the radar image, so that users can visualize the magnitude of displacements from a color code and access historical displacement plots by clicking individual pixels. Pixel sizes vary between 0.25 m to 20 m depending on satellites, with 1 and 3 m being the most common size and precision varying between 1 to 20 mm. Some images are available free of charge from certain space agencies, but the images with the best resolution and allowing the best precision are available commercially.

Some key strengths and attributes of InSAR include its ability to (Pantony et al., 2021):

- Collects dozens of targets on all structures (>50,000 points per square kilometre) on surface areas that can exceed hundreds of km² per SAR image,
- Allows looking back in time to understand movement patterns and establish baselines for construction project sites, and
- Operates remotely from anywhere in the world without the need to visit the site or install any hardware, and to -develop a data density many orders of magnitude greater across a vast territory than any other available technology

Thus, millions of monitoring points can be monitored to a similar precision as a conventional ground survey without any need of field crews or obtaining right of access.

InSAR provides displacement information only in the direction of the radar beam which is oriented at about 50 to 70 degrees from horizontal. So, the true vertical displacement (e.g. ground settlement) of the pixel must be calculated by trigonometry, and the horizontal displacement information is more limited, unless a second radar image is available from a satellite constellation that offers both ascending and descending orbits with respect to the Earth poles. There are however more and more constellations that operate in ascending and descending orbits so that the horizontal information is becoming more available. Another limitation is that the revisit time for each ground surface pixel varies between a few days and up to 14 days or more, although more constellations are planned that will increase the availability of images and improve pixel resolution. Daily revisit times should become available in a quite near future, at least for certain selected areas.

InSAR can provide ground displacement information to a precision as small as 2 mm per year on pixel sizes of 3 x 3 m on images of up to 30 x 50 km with a refreshed display every few days. Additionally, because radar images have been available for a few years already, each site has immediate access to ground displacement information going back a few years, which can be very helpful to provide baseline information for new projects.

Numerous references are available on satellite InSAR, describing its use for urban infrastructure construction projects such as tunnels and bridges, mining surface operations, large ground improvement programs and regional settlement due to groundwater pumping (Ferretti et al., 2007). Pantony et al (2021) provide a good summary of the current capabilities of remote sensing by InSAR.

Satellite InSAR services are available from specialized consultants who provide a turnkey access to the finished product, e.g. a website where all colour-coded displacement information is available over the pre-agreed area of interest on the earth surface (Figure 25-21). Clicking on a given point on the image provides a time series of the historical displacement for that point.

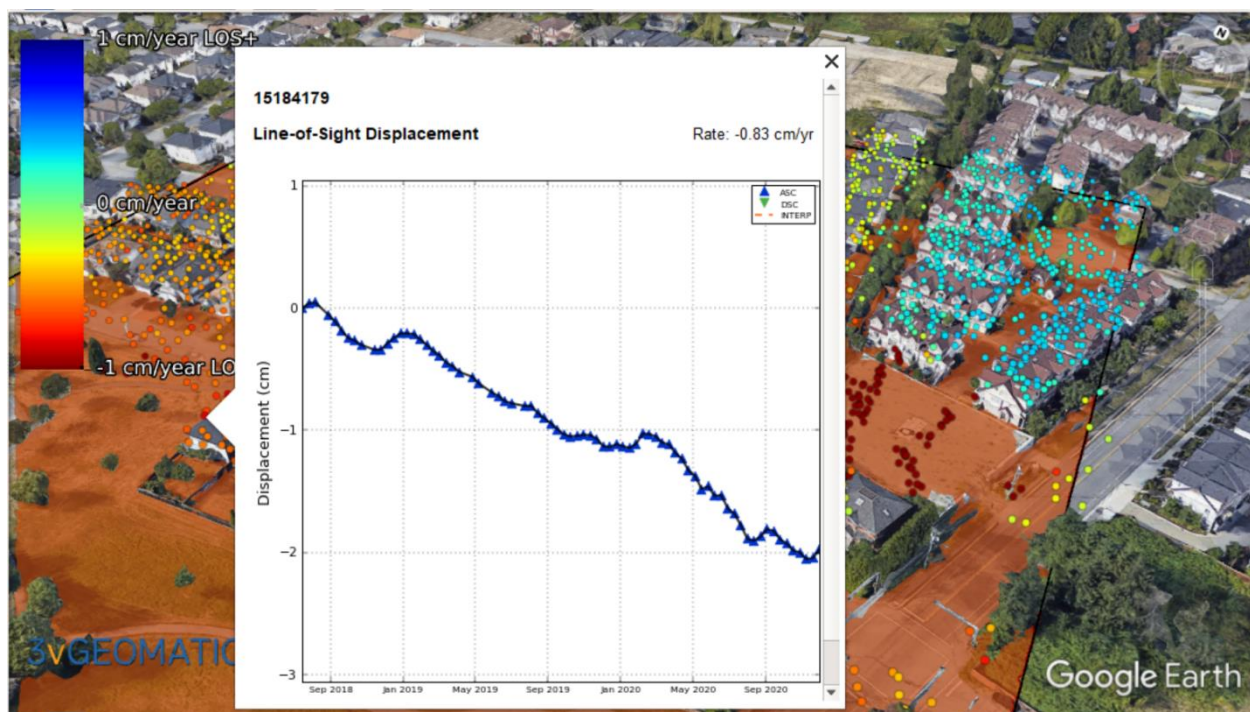
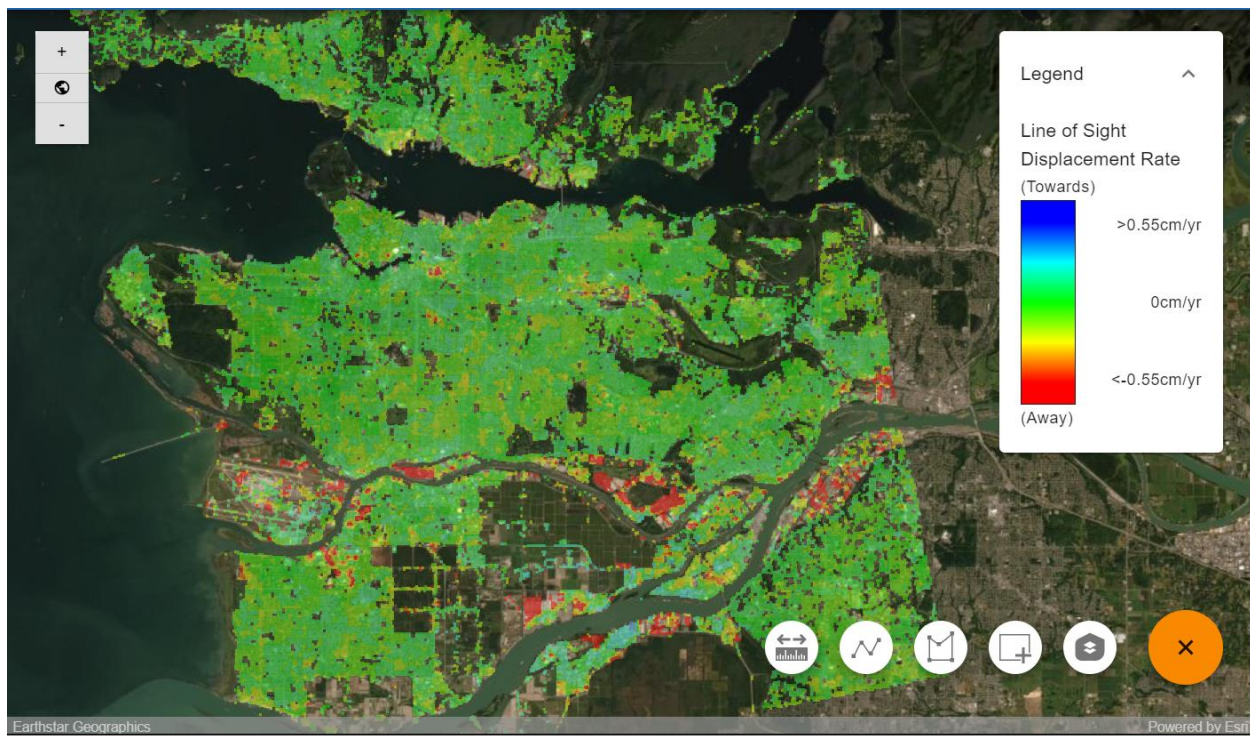


Figure 25-21 Web-based platform showing Vancouver and adjacent cities of the Lower BC Mainland. Largest displacements (mostly vertical settlements) are along the Fraser River and its delta. Bottom image shows a zoom-in to a recently constructed residential neighborhood where displacements of a few centimeters over the past 2 years can be seen on some residences (Pantony et al. 2021).

25.17.6 Ground-based InSAR

Ground-based InSAR uses a similar radar technology as satellite InSAR, but has been adapted to portable units that can be positioned in front of the structure to be monitored. The unit must be mounted on a stable position and the radar scan is conducted automatically on user-defined schedules. Like satellite InSAR, the displacement information obtained is along the line of sight of the radar. Typical applications include permanent or temporary monitoring of dams, rock and soil slopes, bridges, wind turbines, large retaining walls and tall buildings.

Ground-based InSAR can provide displacement information to a precision as small as 0.1 mm on pixel sizes in the range of 0.5 x 5 m at distances from 50 to 5000 m. The precision of 0.1 mm does not account for environmental effects such as humidity and temperature that can affect this number and should be assessed on site for specific situations. Scan durations are as short as 2 minutes or less. It should be noted that ground-based InSAR allows for both static and dynamic monitoring of structures, even to determine natural modes of bridges and similar structures. Andersen et al. (2020) provide a good overview of ground-based InSAR current level of use and acceptance, in addition to the other geodetic and remote sensing technologies discussed in this Section 17. Pieraccini et al. (2019) provide an extensive bibliographic review on the subject.

25.17.7 UAV Photogrammetry

Unmanned Aerial Vehicles (UAV) and photogrammetric processing are often used to inspect difficult to reach locations, documenting cracks and other defects on concrete structures, inspection of dams and levees, mapping joint sets on rock outcrops and other inspection and documentation tasks. UAVs can also be used for deformation monitoring of ground surfaces, embankments and similar structures as reported by Stafford et al. (2019). A UAV consists of a drone that can fly autonomously on user-designed missions. It is equipped with a high-resolution digital camera and a global positioning system (GPS). Aerial images are captured by the UAV and processed using an advanced class of photogrammetric algorithms known as structure-from-motion (SfM). SfM performs a series of automated image matching and calibration algorithms on a set of images with a high degree of overlap that capture the full three-dimensional (3D) scene from a wide array of positions. The product of SfM processing is a dense 3D point cloud of the captured scene, which can be used to monitor deformation when periodic UAV surveys of the same scene are conducted. Using those techniques, Stafford et al. (2019) report being able to measure settlements in the range of 50 to 100 mm on high-fill portions of a highway built on permafrost.

25.18 Measurement of Strong Motions due to Earthquakes

Ground acceleration due to earthquakes is measured by strong motion accelerographs which incorporate a three-dimensional accelerometer and a data acquisition system often called a recorder. The purpose of recording a seismic event is to obtain the peak particle acceleration, expressed in g (gravity acceleration = 9.81 m/s^2), to which the structure has been subjected during the earthquake as well as recording a time history of the seismic event. The stability of structures can be reduced during an earthquake because of excess particle acceleration and it is important to know the magnitude of the exposure to securely put the structure back into service and to post-analyze its stability to improve its resistance to future earthquakes.

Ideally a strong motion accelerograph will be installed in the free field away from the structure of interest, for example embedded in the ground or mounted on a concrete slab directly in contact with the ground, and at least one additional strong motion accelerograph will be installed on the structure at its highest point and at various intermediary positions of interest on the same structure. When more than one strong motion

accelerographs are used on the structure, it can be of interest to synchronize accurately their time stamps using a GPS clock so that time series can be correlated between the different instruments.

Strong motion accelerographs are often classified from Classes A to D, as introduced by the USGS in 2008 for the Advanced National Seismic System (ANSS), a network of strong motion monitoring systems deployed in collaboration with various partners in seismic regions of the United States (U.S. Geological Survey, 2008). Similarly, the Geological Survey of Canada operates a Canada-wide network of over 100 high-gain instruments, called seismographs, and over 60 low-gain or strong motion seismographs called accelerographs, that together make up the Canadian National Seismograph Network (CNSN). The high-gain instruments are used to record weak ground motion from earthquakes that are small and/or distant. Low-gain accelerographs are used to record the strong ground shaking from larger earthquakes at nearby sites likely to cause damage.

Table 25-3 describes the main features of the 4 classes of strong motion systems as defined by U.S. Geological Survey (2008). In this table, the emphasis is on the dynamic range of the combined sensor and data acquisition system which should be broadband, e.g., able to measure seismic motions with wide frequency and amplitude limits generally reaching down to or below site ambient noise. The frequency response includes the long periods and low frequencies needed for global and national seismology purposes. System Class A corresponds to a system at or near the state of the art. System Class B and C can also be used for monitoring of ground accelerations purposes while keeping in mind their reduced specifications may prevent recording events at the low and high end of the range of interest.

Strong-motion Data Acquisition System class	Resolution	Data Acquisition dynamic range (broadband)	
	(μg)	Dynamic range (dB)	A/D Converter Resolution (Bits)
A	<7	>111	≥ 20
B	7–107	87–111	≥ 16 to 20
C	107–1709	63–87	≥ 12 to <16
D	≥ 1709	<63	<12

Table 25-3 Strong-motion systems classification (U.S. Geological Survey, 2008)

Similarly to strong motion accelerations which are defined as accelerations that are easily perceived by humans, lower magnitude accelerations due to traffic on bridges, or ambient vibrations due to smaller tremors can also be measured using accelerometers of higher sensitivity and can be used to evaluate natural frequencies and natural modes of structures such as bridges and tall buildings, which can be an indicator of their structural integrity and how that integrity remains constant or degrades with time and heavy service. Guidelines for strong motion instrumentation of tall buildings are provided by COSMOS (2016).

25.19 Measurement of vibrations

Ground vibrations are discussed in Chapter 17. Vibrations, also called ground velocities, are measured with vibration monitors, which incorporate a triaxial velocity sensor, also called a geophone, and a datalogger. Velocity sensors are sensors similar to accelerometers in their working principle, except that they are more sensitive and are designed for measuring more accurately particle velocities which are expressed in cm/sec,

and the vibration frequency expressed in Hertz. The velocity sensor and the datalogger are either self-contained in a single enclosure or consist in a separate velocity sensor connected with a short cable to the datalogger. The instrument is usually installed directly on the ground, at the toe of the structure to be monitored or it can be installed at a location of interest on the structure. Recent vibration monitor models often include a cellular modem that transmits vibration data to a cloud software.

When installed on the ground, it may be sufficient to put the sensor on the ground and cover it with a sandbag, especially for very temporary installations, or the sensor can have a spike to stick it directly in the ground. Alternatively, it can be mounted on a concrete slab using mounting brackets. Two organizations provide specifications for geophones used in vibration monitors: International Society of Explosive Engineering (ISEE)), Performance Specifications for Blasting Seismographs, 2017 and the German Institute of Standards Deutsches Institut für Normung (DIN), Measurement of vibration immission - Part 1: Vibration meters - Requirements and tests, DIN-45669-1: 2010.

Guidelines are available that provide a correlation between the vibration intensity (e.g., the peak particle velocity in cm/sec) and the vibration frequency (in Hertz) and perception by humans all the way to cracking and/or damage to concrete structures. The two main guidelines are from the US Bureau of Mines Report of Investigation 8507 (Siskind et al., 1980) and the DIN 4150-1, DIN-4150-2 and DIN 4150-3. Acceptable vibration levels can be governed by local or provincial regulations. It can also be a best practice to monitor nuisances to neighboring communities on construction sites to maintain good relations, but also to provide information relative to possible damage claims.

Vibration monitors often have available channels to connect more sensors, such as sound level microphones (to provide sound level in decibels), air overpressure (in kPa) and hydrophones (in kPa) which can all be regulated on construction sites. Dust levels can also be regulated. They are however usually measured with a separate monitoring device.

25.20 Presentation of instrumentation data

The decision to install instruments and develop a monitoring program is a project decision. The monitoring results from an instrument can be of interest to the Owner's engineer, the designers, the contractor and regulators. All monitoring data collected are time stamped and this time stamp forms the link between all data sets collected. Any construction project involves a number of activities, typically consisting of site preparation, excavations, construction of structural members below ground, backfilling and construction of structural members above ground. The ground can be unloaded in preparation for building the structure and then subsequently loaded by building the structure. These construction activities are also time stamped and these time stamps provide the only link to the monitoring data. Because the construction activities deform the ground and change the groundwater levels, it is essential that these construction activities be integrated with the presentation of the monitoring data. Figure 25-22 illustrates this integration of construction activities using the lateral displacement measured in an inclinometer as the twin shallow transit tunnels were constructed in an urban environment.

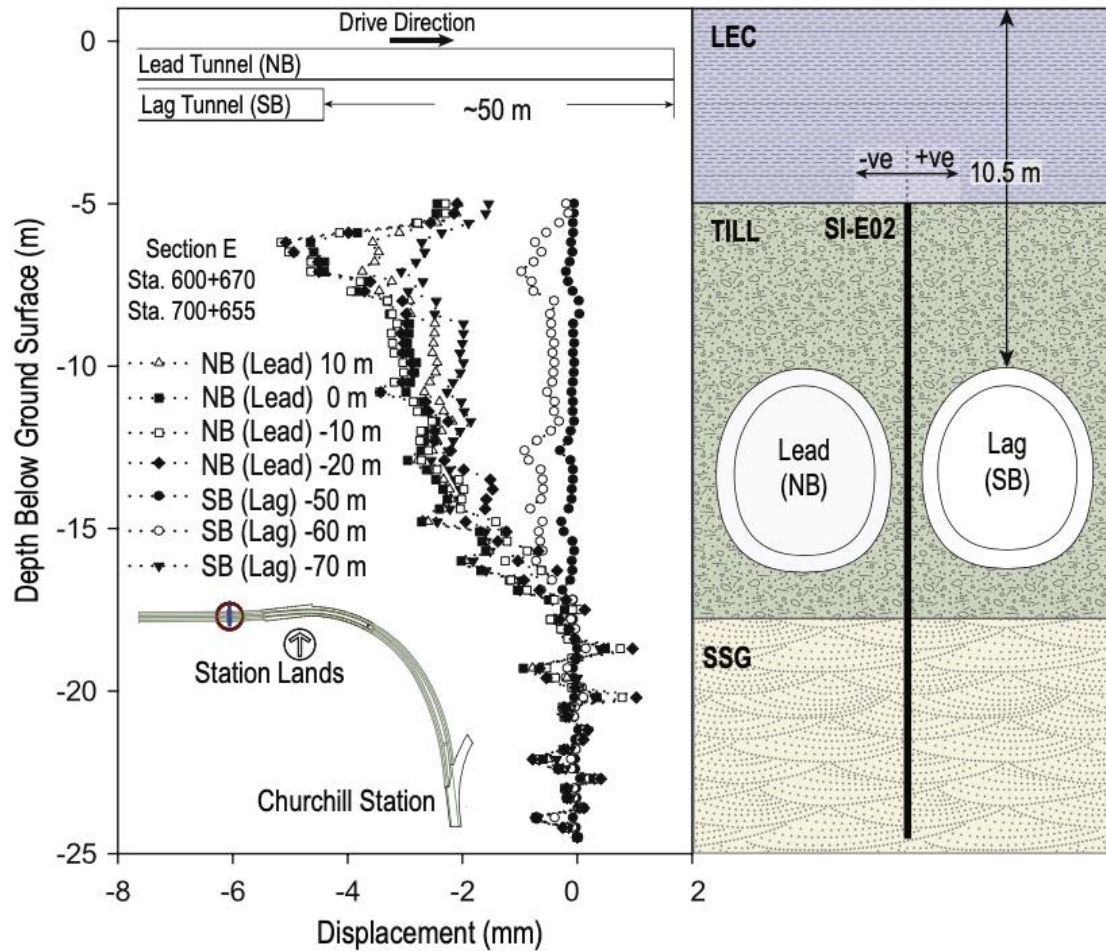


Figure 25-22 Lateral displacements measured between two tunnels in an urban environment. (from Elwood & Martin 2016)

While Figure 25-22 illustrates the impact of construction activities, not all ground responses are caused by construction activities. Figure 25-23 demonstrates the importance of remembering that during unloading the ground can crack and these cracks when filled with rain water can also deform the ground. Figure 25-23 is a reminder that both temperature and rainfall should also be recorded during construction activities and incorporated in the presentation of monitoring data.

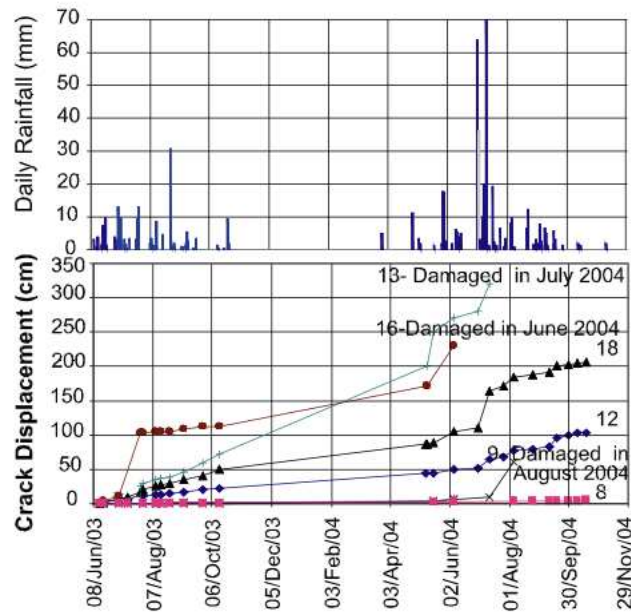


Figure 25-23 Horizontal displacement of crack meters induced by rainfall (from Soe Moe et al 2009)

Despite instrumentation and monitoring being an integral part of geotechnical engineering, the presentation of monitoring data integrated with construction activities and environmental factors remains a challenge for most commercial plotting programs. Any instrumentation and monitoring program must allocate adequate resources so that the monitoring data can be properly presented and interpreted without ambiguity.

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